

Corporate responses to generative AI: early evidence from conference calls

Ning Jia*

School of Economics and Management
Tsinghua University
Beijing, P.R. China

Ningzhong Li

Naveen Jindal School of Management
University of Texas at Dallas
Richardson, TX, USA

Guang Ma

Rutgers Business School
Rutgers University
New Brunswick, NJ, USA

Da Xu

School of Economics and Management
Tsinghua University
Beijing, P.R. China

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*Corresponding author (jian@sem.tsinghua.edu.cn).

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Abstract

We provide early evidence of corporate responses to generative artificial intelligence (GAI) through managerial discussions of GAI in conference calls after ChatGPT's release in November 2022. Following the release, these managerial discussions increase substantially, and most discussions have positive tone and contain action initiatives. The increase is more pronounced for larger firms, growth firms, younger firms, and high-tech industries. The largest increase is observed in the industry of business equipment—computers, software, and electronic equipment, while the utilities, chemicals and allied products, and oil, gas, and coal extraction industries experience the smallest increases. Managers of firms with greater innovation intensity (reflected in patent filings), cybersecurity threats, labor exposure to AI, and customer operations are more likely to increase GAI discussions. The results are primarily driven by positive-tone discussions and those with action initiatives. Analyst discussions of GAI exhibit similar patterns, except for patent filings. Finally, managers' GAI discussions inform investors.

JEL Classification: K4; M4

Keywords: generative AI, ChatGPT, conference call, financial analysts, market reaction

1 Introduction

Disruptive technologies can reshape industries and markets, and generative artificial intelligence (GAI) appears to be no exception. Marked by the release of ChatGPT in November 2022, GAI uses machine learning algorithms and natural language processing to allow users to have human-like conversations with the technology and to create verbal and graphic content by learning from massive amounts of data. ChatGPT quickly gained popularity, attracting over 1 million registered users in just five days and 100 million users in two months (Hu 2023).¹ It has also drawn wide attention to GAI from the public and the business community. Surveyed CEOs believe that GAI will revolutionize how some companies operate (Deloitte 2023).

GAI presents opportunities and threats for firms (McKinsey 2024), which likely vary with industry and firm characteristics. Thus, it is worthwhile to understand how different kinds of firms respond to GAI. We aim to illuminate this by exploring how managerial discussions of GAI in conference calls change following the release of ChatGPT. We assume that managers are aware of GAI-induced opportunities and threats after ChatGPT’s public release and that, if they perceive their firms to be potentially affected, they are likely to talk more about GAI in conference calls. Moreover, if they perceive GAI to represent primarily opportunities (threats) to their firm, they are likely to discuss GAI with a positive (negative) tone.

Conference calls are a preferred setting for this analysis for two key reasons. First, the calls are less formal and regulated (Calomiris et al. 2023) than other forms of disclosure, and thus managers may be franker about their views on GAI than they are in other disclosures.² Consistent

¹ Source: <https://www.reuters.com/technology/chatgpt-sets-record-fastest-growing-user-base-analyst-note-2023-02-01/>.

² Because conference call discussions are less formal and regulated, managers are more likely to discuss GAI strategically, which is a downside of using conference calls. For instance, when managers claim that GAI will benefit their firms, that could be “AI washing” (Barrios et al. 2025). We follow prior research and assume that managers’ discussions in conference calls, on average, reflect their perceptions (e.g., Hassan et al. 2019, 2023; Balakrishnan et al. 2023). We further explore the issue of AI washing in Section 4.3.

with this conjecture, we find that discussions of GAI in 10-Ks are extremely rare in the year following the release of ChatGPT. Second, compared to 10-Ks, conference calls are interactive, so that call participants such as financial analysts can ask questions about GAI even if managers do not voluntarily share their views (Armstrong et al. 2025).

We first provide descriptive evidence on the frequency, intensity, and features (sentiment and whether the discussion includes an action initiative) of managers' GAI discussions in the post-ChatGPT period and compare them with those in the pre-period. We then examine how changes in managerial discussions of GAI (intensity and features) after the launch of ChatGPT vary cross-sectionally with firm and industry characteristics and whether and how these changes inform investors.

Our analyses are based on a sample of 6,192 conference calls hosted by 2,025 public firms between November 30, 2022, and May 27, 2023. To identify GAI discussions, we first search the entire conference call transcript for GAI-related keywords (see Section 3.2), and then, for each keyword hit, we manually read sentences around the keyword to extract all GAI-related sentences. The percentage of conference calls containing GAI discussions increases sharply after the launch of ChatGPT, from 0.34% in the second quarter of 2022 to 9.25% in the second quarter of 2023, indicating that managers promptly respond to this major milestone.

To measure the discussion sentiment, we analyze managers' discussion tone and classify it into either positive, negative, or neutral. Among the post-ChatGPT calls in which managers discuss GAI, 77% have a positive tone, while 10% have a negative tone and 13% have a neutral tone. When discussing GAI, managers may disclose specific action initiatives or merely make broad remarks (e.g., Balakrishnan et al. 2023). We use ChatGPT-4 to identify whether a call's GAI discussions contain GAI-related initiatives and manually verify the results using a random sample

of 100 firms. In 87% of the post-ChatGPT calls in which managers discuss GAI, managers disclose GAI-related initiatives. Coupling managers' positive tone with their specific plans suggests that they generally perceive GAI as an opportunity.

We measure the intensity of managers' GAI discussions in conference calls with the percentage of GAI-related sentences in managers' presentations and answers to analysts' questions. To capture the increase in GAI discussions, we subtract the average of the same percentage measure among conference calls of the same firm in the year before the release of ChatGPT. We find that the increase is more pronounced for firms in high-tech industries, larger firms, growth firms, and younger firms. The largest increase is observed in the Fama-French-12 industry of business equipment-computers, software, and electronic equipment, and the smallest increases are in the utilities, chemicals and allied products, and oil, gas, and coal extraction industries.

We further explore how the change of managers' GAI discussions is associated with several important firm and industry characteristics related to the particular ways GAI is believed to impact firms, including innovation, cybersecurity, labor, and customers (e.g., Deloitte 2023; McKinsey 2023). Specifically, we examine firm innovation intensity, cybersecurity threats, labor exposure to AI, and customer operations. As we elaborate in Section 4.1, it is unclear how these firm and industry characteristics influence managers' GAI discussions. While GAI clearly presents opportunities for corporate innovation (Bilgram and Laarmann 2023; Henry 2023), it is unclear how much managers perceive it as benefiting innovation-intensive firms. GAI could either help less innovative firms catch up to more innovative ones or benefit innovators by further extending their advantages. Regarding cybersecurity, while GAI provides opportunities to increase protection efficiency and improve threat detection (Campbell 2023), it can also introduce new risks by helping attackers launch sophisticated and novel attacks (Irei 2023).

While GAI is generally perceived to offer opportunities to improve firm productivity by substituting for labor and increasing labor productivity (McKinsey 2023; Brynjolfsson et al. 2025), it also can threaten firms. Many employees have concerns about being replaced by GAI, which could harm their psychological welfare, lessen their organizational commitment, and decrease their incentives for human capital investment (Dekker and Schaufeli 1995; Brougham and Haar 2018). Similarly, the impact of GAI on customer operations is also double-edged: while it can improve customer operations (Brynjolfsson et al. 2025), it could also introduce risks, such as providing false information and reducing compassion or empathy (Froehlich 2023).

Empirically, to capture the different aspects of innovation, we measure a firm's innovation intensity by its patent applications, trade secrecy discussions in 10-Ks, and research and development (R&D) expenditure (Gustavo 2011; Fang et al. 2014; Glaeser et al. 2023; Dyer et al. 2024; Glaeser and Lang 2024). Cybersecurity threat is measured with a cyber-risk metric constructed based on the textual analysis of a firm's 10-K filings (Florackis et al. 2023). We capture the degree of labor exposure to AI using the measure of industry-level occupational exposure to AI developed by Felten et al. (2021) and the intensity of customer operations using whether a firm primarily operates in the retail or service sector (Verma 2000; Wang 2024).

We find that, following the release of ChatGPT, firms with more patent applications, greater cybersecurity threats, higher labor exposure to AI, and more customer operations experience a larger increase in managers' GAI discussions in subsequent conference calls. These results imply that managers of those firms perceive their firms to be more affected by GAI. A one-standard-deviation increase in patents (cybersecurity threat, labor exposure to AI) is associated with a higher increase in managers' GAI discussions that accounts for 32% (26%, 37%) of the sample average increase. Industries with more customer operations are associated with a higher increase in

managers' GAI discussions that accounts for 268% of the sample average increase. In contrast, neither trade secrecy discussions nor R&D expenditure are significantly associated with the change of managers' GAI discussions. These results suggest that innovation inputs are insignificantly associated with changes in managers' GAI discussions, and among innovation outputs (patents and trade secrets), only patents show a significant association. This may be because patents are better legally protected, allowing the firm to innovate without fearing appropriation by competitors (Hall et al. 2014). We also examine analysts' perceptions of GAI's impact, as reflected in their discussions during the Q&A sections of calls. We find that their perceptions resemble those of managers, except that changes in their discussions are insignificantly associated with patent filings.

Separately exploring changes of GAI discussions with positive, negative, and neutral sentiments, we find that managers of firms with more patents or greater cybersecurity threats increase their positive discussions of GAI more but not their negative or neutral-tone ones. These findings suggest that managers generally believe that GAI will benefit firms with more patent filings and cybersecurity threats, presumably because it can help them enhance their innovation and cybersecurity. For firms operating in industries with higher labor exposure to AI and more customer operations, managers' views appear to be mixed—they are more likely to increase GAI discussions of all three tones. These findings are consistent with our earlier arguments that GAI could either help or hurt these firms. However, the results for positive-tone discussions are stronger than those for negative-tone ones, suggesting that the perspectives of managers of those firms are more likely to be positive than negative. The results based on analysts' discussions are similar, except that patent filings are insignificantly associated with changes in positive-tone discussions.

We also explore how the four firm and industry characteristics are associated with changes of managers' GAI discussions with and without initiatives. We find that managers at firms with more

patents, greater cybersecurity threats, and more customer operations increase initiative-related discussions of GAI more but not general discussions. Managers of firms in industries with higher labor exposure to AI increase both types of discussions, with stronger results for initiative-related discussions. The results based on analysts' discussions are similar, except that patents are insignificantly associated with changes of initiative-related discussions.

Next we find that larger increases in GAI discussions (intensity, discussions with positive tone, and initiative-related discussions) are associated with stronger market reactions and higher abnormal trading volume, and firms with higher institutional ownership primarily drive the responses. Furthermore, the results for initiative-related discussions strengthen when the discussions are more specific and certain. This evidence suggests that managers' GAI discussions inform investors.

Lastly, we explore how the four firm and industry characteristics are associated with the market reaction to the launch of ChatGPT, which reflects investors' perceptions of GAI's impacts. We find that the market reaction is positively and significantly associated with patent filings and cybersecurity threats, which is consistent with the findings based on managers' GAI discussions. However, unlike the findings from manager discussions, the market reaction is negatively and significantly associated with R&D expenditure and trade secrets, and is insignificantly associated with labor exposure to AI or customer operations. These results suggest that investors' perceptions of GAI can differ from those of managers (and analysts). This justifies separately examining managers' perceptions (as well as analysts') on top of research based on market reactions (e.g., Eisfeldt et al. 2023).

A potential concern is that managers' GAI discussions in conference calls may reflect "AI washing"—making false, misleading, or exaggerated statements about whether and how firms are

using or developing AI systems (e.g., Barrios et al. 2025). For example, some GAI initiatives discussed may be superficial from an economic perspective. We provide some analyses to mitigate this concern, but acknowledge that we cannot rule out this concern and leave further exploration of this issue for future research.

Our paper adds to the nascent literature on the corporate impacts of GAI. Given the revolutionary nature of ChatGPT, it is unsurprising that scholars as well as practitioners have tried to understand its impact. Concurrent studies have examined its overall valuation impact (e.g., Eisfeldt et al. 2023) and productivity impact in specific settings (e.g., Gilardi et al. 2023; Yilmaz et al. 2023; Brynjolfsson et al. 2025).³ We add to this literature by providing systematic evidence on how and under what circumstances managers discuss GAI. We provide rich, detailed evidence on the frequency, intensity, and features of managers' GAI discussions in the period right after ChatGPT's public launch and compare them with the pre-period. We also examine how changes in GAI discussions vary with industry membership and fundamental firm characteristics. More importantly, we show how the discussion changes are associated with firm and industry characteristics related to the major channels through which GAI is generally perceived to impact firms. Our evidence based on analysts' GAI discussions complements our findings and inferences about corporate responses.

Overall, our study provides early insights into how managers, analysts, investors, and other stakeholders perceive GAI's potential influences. Our findings suggest that perceptions of GAI may vary among stakeholders. Once more data on firms' actual use of GAI become available, future research could assess the real impacts of GAI and determine whether they align with the currently documented perceptions.

³ We review the related literature in Section 2.2.

2 Background, literature review, and conceptual framework

2.1 *Background of generative AI*

Adopting AI for human activities has a long history. In 1954 IBM made the first public demonstration of a machine translation system (Zhang et al. 2023). However, it was not until November 2022 that AI captured the global attention with the introduction a generative AI product, ChatGPT, by OpenAI. This GAI application has emerged from foundation models—large-scale, deep learning models trained on massive structured and unstructured datasets (text, voice, etc.) that cover a wide range of topics. Unlike traditional AI tools designed to summarize patterns and make predictions, GAI can create content for a broad range of uses with little fine-tuning required for each case. For example, ChatGPT has been used to translate text and to create protein sequences. ChatGPT capabilities are accessible to all, including people with little computer literacy and technical background.

Due to its wide applications and low usage barrier, GAI is widely perceived to significantly affect society and economy. It has the potential to reshape knowledge work in industries and business functions across the economy. Across such functions as sales and marketing, product development, and innovation activities, it is poised to transform roles and boost performance (McKinsey 2023). As a result, the release of ChatGPT has drawn public attention to this product and to GAI. Figure 1 presents the trend of worldwide Google search interest for the phrase “ChatGPT,” “GPT,” and “generative AI/generative artificial intelligence.” The search interest surged sharply after the release of ChatGPT and continued to grow until the end of our examination window, June 2023.

GAI has been attracting the interest of investors and corporations. The funding for GAI has been growing rapidly, reaching \$12 billion in the first five months of 2023 alone (McKinsey 2023).

More companies are investing in their own GAI products. Among the major players, Microsoft has reportedly invested \$10 billion in OpenAI and ChatGPT. In December 2023, Google introduced its GAI product, Gemini. Despite the fast development of GAI, this technology is not without challenges and limitations. For example, the accuracy of the results can be questionable. GAI tools can generate inaccurate information in response to a user question and have no built-in mechanism to signal this to the user or challenge the result. Another concern is systemic biases in results as GAI tools may be trained on data that may include unwanted biases (Chilton 2023).

2.2 Literature

Given the significance of GAI, the emerging literature has begun to examine its impacts on businesses, especially following the launch of ChatGPT. Eisfeldt et al. (2023) focus on the valuation implications of a particular type of firm exposure to GAI—workforce exposure. They measure this exposure by analyzing job tasks and their susceptibility to AI-driven automation or enhancement, creating an index that quantifies exposure across a company’s workforce. Using a portfolio strategy, they find that higher-exposure firms earn higher excess returns compared to lower-exposure ones. They conclude that the release of ChatGPT is generally viewed by investors as good news for more exposed firms. However, they also find that investor perceptions vary across and within industries.

A stream of research examines the productivity impact of GAI, usually focusing on how GAI performs compared to humans and how it can improve labor productivity in a specific setting (e.g., Kreitmeir and Raschky 2023; Noy and Zhang 2023; Eulerich et al. 2024). Bertomeu et al. (2025) examine how GAI facilitate financial analysts’ information processing using the Italian ban on ChatGPT as an exogenous shock. They find that the ban diminishes analyst forecast accuracy and market efficiency. Yilmaz et al. (2023) use Google’s introduction of neural network-based

translation (GNNT), a type of GAI-based application, as a natural experiment to examine the substitution of human translators by AI. They find that the introduction of GNNT reduces the number of human translation transactions. Brynjolfsson et al. (2025) examine the impact of GAI-based tools on labor productivity in customer support activities. They find that customer support agents using AI tools become more productive, as reflected in the number of customer issues resolved per hour. Using a sample of tweets, Gilardi et al. (2023) demonstrate that GAI outperforms crowd-workers for several text-annotation tasks.

2.3 Conceptual framework

We aim to add to the understanding of GAI's impacts by examining how firms responded to GAI based on managers' GAI discussions in conference calls. Research has shown that conference calls provide additional insight beyond traditional disclosures, with much of their informativeness attributed to their interactive and sometimes unscripted nature (Frankel et al. 1999; Matsumoto et al. 2011; Lee 2016).⁴ Therefore, the literature has used conference calls, and corporate disclosure broadly, to measure managerial perceptions of various issues, such as corporate culture (Li et al. 2021), political risk (Hassan et al. 2019), Brexit uncertainty (Hassan et al. 2024a), epidemic exposure (Hassan et al. 2023), and cybersecurity risk (Jamilov et al. 2021), assuming that managers' discussions in conference calls, on average, reflect their views and corporate actions. For example, Balakrishnan et al. (2023) use conference calls to measure a firm's exposure to diversity issues and whether the firm undertakes initiatives to advance its diversity-related efforts. Hassan et al. (2024b) use conference calls held by listed firms around the world to measure the amount of risk that managers and investors perceive in each country.

⁴ The unscripted dialogue in the Q&A sessions often reveals managers' genuine perspectives on key issues (Li et al. 2021).

We use a similar framework, assuming that managers' GAI discussions reflect their firms' responses to GAI. The responses in managers' discussions may include their real actions taken, their plans, or their views without any actions or plans. We assume that these responses stem from managers' perceptions of how GAI may affect their firms, which may vary with firm and industry characteristics, such as innovation intensity and cybersecurity threats. If managers perceive their firms to be potentially affected by GAI, they are likely to talk more about GAI. If managers perceive GAI to represent primarily opportunities (threats) to their firm, they are likely to discuss GAI with a positive (negative) tone.

One concern is that managers' discussions of GAI during conference calls may involve AI washing. Several factors help mitigate this concern. Litigation risks and the need to adhere to securities regulations discourage managers from providing materially misleading information (Heitzman et al. 2010; Huang et al. 2020). Additionally, market discipline and reputation concern further deter misrepresentation (Healy and Palepu 2001). The sophistication of analysts and institutional investors, who are skilled in identifying inconsistencies, adds additional scrutiny (Lang and Lundholm 1993). Finally, as discussed above, the real-time nature of conference calls also helps ensure accountability. Managers must answer questions immediately, and follow-up inquiries can expose their evasions. This dynamic impedes managers from biasing or exaggerating information. We provide further discussion of AI washing in Section 4.3.

3 Sample selection, measurement of GAI discussions, and descriptive evidence

3.1 Sample selection

Our sample comprises publicly listed companies on the NYSE and Nasdaq, with financial data accessible from Compustat. Initially, we consider the universe of 8,065 companies in Compustat, all of which have reported positive total assets for the fiscal year ending between

November 29, 2021, and November 30, 2022 (the public launch date of ChatGPT). After requiring availability of firm variables, our sample shrinks to 2,869 firms. We search for firms' conference call transcripts in Capital IQ Transcript Database and retrieve 6,772 transcripts for our sample firms, spanning the period of November 30, 2022, to May 27, 2023. After excluding transcripts without conference call variables, we are left with a sample of 6,192 transcripts for 2,025 firms. For stock return analysis, we further require the availability of stock price data from Center for Research in Security Prices (CRSP). The sample selection procedures are detailed in Appendix A.

3.2 *Measurement of GAI discussions*

To identify GAI discussions in conference calls, we first compile a list of GAI-related keywords along three dimensions—vendor, product, and technology. We posit that managers are unlikely to discuss GAI without mentioning any of these three dimensions. For vendor and product-related keywords, we identify leading GAI vendors by referencing several commercial rankings, and for each vendor, we obtain information about its product offerings from its official website.⁵ For technology keywords, we first extract a preliminary list of keywords from Zhang et al. (2023), a computer science survey paper on GAI technology. We then ask three technology experts to independently select the keywords that they perceive to be highly relevant to GAI.⁶ We retain keywords that at least two choose. The final keyword list is provided in Appendix B.

For each conference call transcript, we search for the GAI-related keywords in the entire transcript and identify sentences that contain the keywords.⁷ We then manually read sentences

⁵ We referenced several commercial rankings, including eWeek's *Generative AI Companies: Top 12 Leaders* (<https://www.eweek.com/artificial-intelligence/generative-ai-companies/>) and CB Insights' *Gen AI: The Most Promising Generative Artificial Intelligence Startups of 2023* (<https://www.cbinsights.com/research/generative-ai-top-startups-2023/>).

⁶ Two are computer science professors at a leading university, whose research focuses on AI, and one is a chief technology officer of an AI firm.

⁷ Of all 10-Ks of our sample firms released in the year after ChatGPT's public launch, only 0.97% contain GAI-related keywords described in Section 3.2. After reading these 10-Ks, we find that most keyword-related sentences are not

around them to identify all GAI discussions. Hence, our final GAI-related sentences consist of both the keyword sentences and the context. We provide examples of these discussions in Appendix C.

3.2.1 *GAI discussion intensity*

To measure the intensity of managers’ GAI discussions, we compute the percentage of GAI-related sentences relative to the aggregate sentence count in management discourses—the presentations and answers to analysts’ questions and denote this variable as *GAI%*. To capture the change of GAI discussions following ChatGPT’s release, we subtract the average of *GAI%* among conference calls of the same firm between November 30, 2021, and November 29, 2022, and denote the change as $\Delta GAI\%$. We use a similar approach to identify analysts’ GAI discussions based on analysts’ questions during the call Q&A. The corresponding variables are denoted as *GAI%-Analyst* and $\Delta GAI\%-Analyst$.

3.2.2 *GAI discussion sentiment*

To measure the tone of managers’ GAI discussions, we classify the net sentiment of the discussions in each conference call as positive, negative, and neutral. We use the lexicon introduced by Loughran and McDonald (2011) to ascertain the net sentiment of a call’s GAI discussions. A call with GAI discussions is categorized as having positive GAI discussions if the net sentiment is positive (above zero), neutral if the net sentiment equals zero, and negative if the net sentiment is negative (below zero). We present one example of each kind of discussion in Appendix C Panel A.

After measuring the net sentiment of a call’s GAI discussions, we decompose *GAI%* into *GAI%-Pos*, *GAI%-Neg*, and *GAI%-Neu*. *GAI%-Pos* equals *GAI%* if the net sentiment of GAI

about GAI. This is not an issue in our search of conference calls; the machine-identified GAI discussions based on keyword search are verified manually to ensure the discussions are truly about GAI.

discussions is positive, and zero otherwise. *GAI%-Neg* and *GAI%-Neu* are defined similarly. By construction, *GAI%* is the sum of the three sentiment-based measures. We use this holistic approach (rather than measuring each sentence’s sentiment) because it better reflects managers’ *overall* perception of GAI. We then calculate the change of *GAI%-Pos* by subtracting the average *GAI%-Pos* of the same firm in the pre-ChatGPT period (between November 30, 2021, and November 29, 2022) and denote the change as $\Delta GAI\%-Pos$. We define $\Delta GAI\%-Neg$ and $\Delta GAI\%-Neu$ similarly. We use a similar approach to construct the corresponding variables for analysts and denote them by adding the suffix *-Analyst* to the manager variables.

3.2.3 Initiative-related versus non-initiative-related GAI discussions

We use ChatGPT-4 to identify whether a call’s GAI discussions contain GAI-related initiatives and manually verify the results using a random sample of 100 firms.⁸ ChatGPT-4 performs fairly well in distinguishing the two kinds of discussions, achieving a machine-human consistency rate of 86%.⁹ Appendix C Panel B provides one example of each type. After the classification, we follow a similar procedure as in Section 3.2.1 to calculate GAI discussions with initiatives, *GAI%-Initiative*, and discussions without, *GAI%-NonInitiative*, and their changes, $\Delta GAI\%-Initiative$ and $\Delta GAI\%-NonInitiative$. We use a similar approach to construct the corresponding variables for analysts (whether they ask about GAI initiatives) and denote them by adding the suffix *-Analyst* to the manager variables.

3.3 Summary statistics

Table 1 presents the summary statistics for our sample. Panel A reports the statistics for the

⁸ We use the following prompt: “Does the text involve real initiatives or plans in generative AI? Answer with 1 for Yes, 0 for No. Text: { }”

⁹ We validate ChatGPT’s classification with human classification by two research assistants, each examining a randomly selected subsample of GAI discussions of 100 firms. The research assistants are provided the same question as the ChatGPT prompt. The accuracy rate of ChatGPT coding benchmarked against human coding is 84% and 88% respectively for two research assistants.

GAI discussion variables, which we will discuss in Section 3.4.1 below. Panel B reports the statistics for firm characteristics. The average total assets of sample firms is 15.852 billion dollars. The average firm is 27 years old and has a market-to-book ratio of 6.449, 13 analysts following it, and 76% institutional ownership. About half (48.6%) of firms are in high-tech industries. Panel B also reports the statistics on firm characteristics related to innovation, cybersecurity, labor, and customer as well as market reaction-related variables, which we will discuss below.

3.4 Descriptive evidence on managers' GAI discussions

3.4.1 Frequency, intensity, features, and changes

Panel A of Table 1 shows the average intensity and features of managers' GAI discussions and their changes. Regarding discussion intensity, the mean of $GAI\%$ is 0.316, indicating that in the post-ChatGPT period 0.316% of managers' discussions in conference calls relate to GAI. The mean of $\Delta GAI\%$ is 0.306, which means relative to the benchmark period of one year before the launch of ChatGPT, GAI discussions increase by 0.306 percentage points on average. The difference between $GAI\%$ and $\Delta GAI\%$ indicates that the average value of $GAI\%$ in the pre-ChatGPT period is only 0.01 percentage points. Thus, managers rarely discussed GAI before ChatGPT, and their discussions increase sharply following its public launch.

To provide further insight on the time-series trend of managers' GAI discussions, in Figure 2A, we present the quarterly distribution of the percentage of conference calls that contain GAI discussions over the fourth quarter of 2021 to the second quarter of 2023. Prior to 2023, on average only less than 1% of calls contained GAI discussions. The percentage surged sharply to around 9% in the second quarter of 2023, consistent with the statistics reported in Table 1. Figure 2B presents the quarterly distribution of $GAI\%$. Its increasing trend resembles that in Figure 2A.

Regarding the sentiment of managers' GAI discussions, Panel A of Table 1 reports that the

means of *GAI%-Pos*, *GAI%-Neg*, and *GAI%-Neu* are 0.263, 0.030, and 0.019, respectively. The average values of *GAI%-Initiative* and *GAI%-NonInitiative* are 0.305 and 0.012, respectively. As the mean of *GAI%* is 0.316, these statistics suggest that managers' GAI discussions mostly have positive net sentiment and contain action initiatives.¹⁰ This indicates that managers of firms likely affected by GAI generally view it as presenting opportunities to their firms and are inclined to capitalize on such opportunities by implementing GAI-related actions.

For comparison, we also report the descriptive statistics for analysts' GAI discussions in conference calls in Table 1 Panel A, which show similar patterns and features as managers' discussions. Specifically, analysts rarely discuss GAI in the pre-ChatGPT period, and their discussions increase substantially following the release of ChatGPT. In the post-period, when analysts discuss GAI, they typically express positive views and ask about action initiatives. Thus, it seems that analysts' perceptions of GAI, on average, resemble those of managers. Consistent with this observation, the correlation between *GAI%* and *GAI%-Analyst* is high (0.64), so is that between $\Delta GAI\%$ and $\Delta GAI\%-Analyst$ (also 0.64).

3.4.2 Cross-industry variations

We explore the cross-industry (Fama-French-12) variation of changes of managers' GAI discussions (intensity and features) in Figure 3. As managers rarely discuss GAI in the pre-ChatGPT period, the figures based on the level of managers' GAI discussions in the post- period are very similar and thus omitted. Figure 3A presents the variation for changes of discussion intensity ($\Delta GAI\%$). While on average managers' GAI discussions increase (Table 1), there are

¹⁰ The average net sentiment score for GAI discussions is 1.55%. In an untabulated analysis, we calculate the average net sentiment scores for non-GAI discussions, ESG-related discussions, and competition-related discussions, using the dictionaries of Briscoe-Tran (2021) and Li et al. (2013). The averages are 1.08%, 1.13%, and 1.10%, respectively, all significantly lower than that of GAI discussions. Therefore, the positive sentiments we document for GAI discussions are not generic to all topics.

significant variations across industries. The Fama-French-12 industry of business equipment—computers, software, and electronic equipment experiences the largest increase in managers’ GAI discussions, and the increase is much higher than that in any other industry. The mean of $\Delta GAI\%$ in the business equipment industry is 1.067, while those of the other industries are all below 0.2. The three industries with the least increase in managers’ GAI discussions are utilities, chemicals and allied products, and oil, gas, and coal extraction with the means of $\Delta GAI\%$ 0, 0, and 0.003, respectively. Managers of firms in these industries almost never discuss GAI before or after ChatGPT’s arrival. This cross-industry pattern is somewhat expected because high-tech industries such as business equipment are more exposed to automation and optimization through GAI (McKinsey 2024). In contrast, industries like utilities rely more on capital investments, physical processes, and infrastructure and thus are probably viewed by managers as less affected by GAI.

Figure 3B separately presents the changes of managers’ GAI discussions with different sentiment. For industries with nontrivial discussion changes, the relative changes of positive-, negative-, and neutral-tone discussions are generally consistent with the general pattern reported in Panel A of Table 1—the increased discussions are predominantly driven by positive-tone discussions while negative- and neutral-tone discussions also increase, with two exceptions. First, the increase in positive-tone discussions is smaller than those in negative- and neutral-tone discussions in the consumer durables industry. Second, there are no changes of negative-tone discussions in two industries: manufacturing and telephone and television transmission. It seems in managers’ views, the influence of GAI in the consumer durables industry is more likely to be negative relative to an average industry, and it is more likely to be positive in the manufacturing and telephone and television transmission industries.

Figure 3C separately presents changes of managers’ GAI discussions with and without

initiatives. For industries with nontrivial discussion changes, the relative changes of initiative- versus non-initiative-related discussions are generally consistent with the pattern reported in Panel A of Table 1—the increased discussions are predominantly driven by discussions of initiatives, with notable exception of the telephone and television transmission industry, which has no discussions of initiatives.

We use Fama-French 12 industries in Figure 3 to save figure space. To provide a more granular comparison, we categorize two-digit SIC industries into high-tech (SIC codes 28, 25, 36, 73, and 87) and non-high-tech ones (the other SIC codes), following Kimbrough and Louis (2011) and De Franco et al. (2023). Table 2 Panel A reports that the increases of managers' GAI discussions, overall intensity as well as features (except the ones without initiatives), are all significantly higher in high-tech industries. This pattern is consistent with Figure 3.¹¹

3.4.3 Variations based on fundamental firm characteristics

In Panels B to D of Table 2, we further compare firms with different fundamental characteristics, including size, age, and growth potential (market-to-book ratio). The sample partition is based on the sample median of each partitioning variable, namely, *Size*, *Firm Age*, and *Market to Book*, which are defined in Appendix D. We find that the increases in managers' GAI discussions, overall as well as those with different features, are generally more pronounced for larger firms, growth firms, and younger firms.¹²

These patterns are consistent with economic intuition because these firms arguably have more resources, cost advantages, incentives, or flexibility to incorporate the new GAI technology. For example, larger firms generally have more resources; they also have strong incentives to invest in

¹¹ For comparison, we present in Figure IA-1 and Table IA-1 in the internet appendix a similar set of comparisons for analysts' GAI discussions. These patterns are largely consistent with those of managers' discussions.

¹² There are two exceptions: changes of discussions without initiatives and those with neutral tones do not vary significantly with firm size.

new technologies to achieve long-term benefits. Younger firms and growth firms emphasize innovation and growth; they are also more agile and willing to adopt to new technologies.¹³

4 Analysis of the economic determinants of GAI discussions

4.1 Conceptual discussion and measurement of key variables

The resource-based view of a firm posits that heterogeneity in firm performance is rooted in variation in resources and the ability to achieve and sustain a competitive advantage (Barney 1991; Wernerfelt 1984). From this perspective, we posit that GAI will likely impact firm performance through its impacts on corporate resources and capabilities. Based on economic intuitions, anecdotes, and practitioner views (e.g., Deloitte 2023; McKinsey 2023), we explore four key dimensions—innovation, cybersecurity, labor, and customers. We provide in Appendix C Panel C examples of GAI discussion for the four dimensions we examine. We posit that, if managers perceive their firms to be potentially affected in a given dimension, they are likely to increase GAI discussions and that if managers perceive GAI to represent primarily opportunities (threats) to their firm in a given dimension, they are likely to discuss GAI with a positive (negative) tone.

4.1.1 Innovation

GAI can benefit corporate innovation (Deloitte 2023; McKinsey 2023). For example, it can accelerate the process of exploration, facilitate prototyping at lower costs, help researchers brainstorm, and automate routine tasks such as data analysis, report generation, and content creation (Reed 2022; Bilgram and Laarmann 2023; Henry 2023). Empirically, we explore the association between firm innovation intensity and changes of managers' GAI discussions following the release of ChatGPT. It is unclear whether GAI's innovation benefits are more

¹³ For comparison, we present in Panels B to D of Table IA-1 in the internet appendix a similar set of tests for analysts' GAI discussions. We find that changes in analysts' GAI discussions are generally more pronounced for young firms and growth firms, while the changes do not vary significantly with firm size.

valuable for firms with greater or less innovation. On the one hand, GAI could help less innovative firms to catch up to more innovative ones. On the other hand, GAI may be more beneficial for innovators by further extending their innovation advantages.

Glaeser and Lang (2024) note that no single dominant empirical measure captures corporate innovation, and each measure has its advantages and limitations. Therefore, we employ two commonly used output-based measures—patents (*Patents*) and trade secrets (*Trade Secret*)—and one input-based measure—R&D expenditure (*R&D*) (e.g., Gustavo 2011; Glaeser et al. 2020; Glaeser and Landsman 2021).¹⁴ While innovation inputs such as R&D expenditures better capture firms’ investment in innovation, the outcomes of these investments could be uncertain. Among the innovation outputs, patents are better legally protected than trade secrets (Hall et al. 2014; Glaeser and Lang 2024).

To calculate *Patents*, we count the total number of patent applications submitted to the United States Patent and Trademark Office (USPTO) by a firm and its subsidiaries in the past three calendar years. To address the skewness of the data, we use the inverse hyperbolic sine transformation, which adjusts for skewness and is well defined for nonpositive values (e.g., Burbidge et al. 1988; Glaeser and Omartian 2022). We follow Glaeser (2018) and use trade secret discussions in 10-K filings to measure the existence of trade secrets. Specifically, *Trade Secret* is defined as a dummy variable that equals 1 if the firms’ 10-K filings in the past three years mention “trade secret” or “trade secrecy.”¹⁵ *R&D* is measured as R&D expenditure over total assets. Missing R&D expenditures are coded as zero. To address the issue that firms with missing R&D

¹⁴ While filing for patents is one way to protect a firm’s innovation output, many firms may choose not to file patents due to proprietary information concerns (Hall et al. 2014). Another way to protect innovation output is through trade secrets, which is generally defined as “information that derives future economic value from not being appropriable by competitors (e.g., unpatented innovations).” (Glaeser 2018 pp. 163–164).

¹⁵ We thank Professor Stephen Glaeser for making the trade secret data available at <https://www.stephenglaeser.com/data>.

expenditures could have significant R&D investments (Koh and Reeb 2015), we also include an indicator variable for missing R&D expenditures, denoted as *Miss R&D*. Table 1 Panel B reports that an average firm has 32 patent filings in a year (*Patents*, untransformed). The mean of *Trade Secret* is 0.829, suggesting that 82.9% of firm-years discuss trade secrecy in the 10-Ks in the past three years. *R&D* has a mean of 0.064 and a median of 0.012; 33.1% of firms have missing R&D expenditure data.

4.1.2 Cybersecurity threats

It is unclear whether GAI would reduce or exacerbate a firm's cybersecurity threats because it can lead to both benefits and risks in cybersecurity (Campbell 2023; Irei 2023). Regarding the benefits, GAI can increase protection efficiency, improve threat detection, automate responses, and continuously monitor (Campbell 2023). However, GAI can also introduce new cybersecurity risks by helping attackers launch sophisticated and novel attacks. For example, ChatGPT can be used to write malware and help hackers build their skills (Irei 2023).¹⁶ In addition, employees using GAI tools like ChatGPT could put sensitive corporate information at risk.¹⁷ For these reasons, it is unclear whether firms with higher or lower cybersecurity threats are affected more by GAI regarding their cybersecurity and in which direction.

We assess cybersecurity threats by evaluating the likelihood that a firm might face cyberattacks using an approach based on Florackis et al. (2023). Specifically, we compare a firm's description of its cybersecurity vulnerabilities in the Item 1a. Risk Factors section of its 10-K filings against the characteristics of all documented past cyberattacks of firms. A higher overlap

¹⁶ For example, malicious hackers could request ChatGPT to create code for penetration testing and then modify and repurpose it for use in cyber-attacks (Irei 2023).

¹⁷ For example, in April 2023, Samsung witnessed three separate leaks of sensitive company data with ChatGPT, including computer codes and meeting records. (source: <https://www.foxbusiness.com/technology/samsung-employees-reportedly-leaked-sensitive-info-chatgpt-accident>)

(similarity) between the focal firm’s cybersecurity risk disclosure and traits of actual past cyberattacks suggests a higher possibility of *real* threats.

We gather information on cybersecurity incidents from the Privacy Rights Clearinghouse’s (PRC) Data Breach Chronology Database, focusing exclusively on incidents related to cyberattacks or hacking. The PRC database provides extensive details on each incident, such as the nature of the information compromised and the incident’s magnitude. We compile the descriptions of these incidents to create a reference point for evaluating real cybersecurity threats. Next we extract statements related to cybersecurity from the section Item 1a. Risk Factors of a firm’s 10-K filings. We identify relevant disclosures by searching for specific keywords, as suggested by Florackis et al. (2023), such as “cyberattack,” “data breaches,” and “data theft.” We collect all sentences containing these keywords as cybersecurity disclosures for each firm-year. Lastly, we calculate the cosine similarity between the text of the firm’s disclosures and the descriptions of real past cyber incidents in the PRC dataset, denoted as *CS Threat*. A higher value suggests that the company’s disclosures closely match the details of actual cyberattacks, indicating a higher vulnerability. Table 1 Panel B reports that *CS Threat* has a mean and a median of 0.691 and 0.695, respectively, suggesting that 69.1% of an average firm’s cybersecurity-threat disclosures in 10-K filings resemble the actual incidents from the PRC Data Breach Chronology Database.

4.1.3 Labor exposure to AI

GAI is widely perceived to have great potential to substitute for labor and increase labor productivity (McKinsey 2023; Yilmaz et al. 2023; Boyaci et al. 2024; Brynjolfsson et al. 2025). However, what is less obvious is that GAI might also hurt labor productivity and firm performance. When GAI is developing quickly, many employees have increasing concerns about job insecurity,

which could hurt their psychological health, lessen organizational commitment, and decrease their human capital investment (Dekker and Schaufeli 1995; Brougham and Haar 2018). To investigate how managers view the influence of GAI on firms through its effect on labor, we examine how managers of firms whose labor is more affected by AI (higher labor exposure to AI) change their GAI-related discussions in conference calls relative to other firms after ChatGPT’s release. Given the expected impact of GAI on labor, we expect that managers of firms with higher AI exposure are more likely to increase GAI-related discussions. However, based on the two-sided arguments above, it remains unclear whether their perspectives tend to be positive or negative and whether they are more likely to discuss concrete actions.

We measure a firm’s labor force exposure to AI using the AI Industry Exposure (AIIE) developed by Felten et al. (2021).¹⁸ Felten et al. (2021) identify a set of common and well-developed applications of AI, relate these applications to abilities in workplace activities of each occupation, and develop a measure of an occupation’s exposure to AI technologies, which they call AI Occupational Exposure (AIOE). They validate their measure in several ways, for example, by showing that AIOE is positively associated with use of AI skills within an occupation based on the job postings. Due to the lack of firm-level occupation data for the construction of a firm-level AI exposure measure, we rely on their measure, which is constructed by aggregating AIOEs across occupations within an industry (AIIE). This means that empirically we focus on between-industry variation of labor exposure to AI. We denote this measure as *AIIE*. The mean of *AIIE* in our sample is 0.642, suggesting that on average 64.2% occupational abilities relate to AI applications at the industry level (Table 1 Panel B).

4.1.4 Customer operations

¹⁸ The measure is available at <https://github.com/AIOE-Data/AIOE>. We thank the authors for making the data available.

While GAI is generally believed to significantly impact customer operations, it is unclear whether it will always improve customer operations. On one hand, GAI may revolutionize the entire customer operations function (BCG 2023; Siggelkow and Terwiesch 2023; Brynjolfsson et al. 2025). On the other hand, there are also risks associated with using GAI in customer services. For instance, it can provide false or biased information, alienate different customer segments, and lack the compassion and empathy of a live customer agent, which could harm customer operations (BCG 2023; Froehlich 2023). We anticipate that managers of firms with greater customer operations will likely intensify their discussions about GAI more. Nevertheless, it remains uncertain whether their viewpoints are predominantly positive or negative and whether they are more likely to discuss actions.

Due to the lack of individual customer data, we also rely on the industry-level variation to measure customer operations.¹⁹ We posit that firms in the retail and services industries entail greater customer operations than those in other industries (Verma 2000; Wang 2024). We identify retail and services industries based on two-digit SIC industry codes. Specifically, we classify the retail industry by two-digit SIC codes 52–59 and the service industry by two-digit SIC codes 72, 73, and 78 (Cole and Jones 2004). We define a variable *Customer* that equals 1 if a firm is in a retail or services industry and 0 otherwise. Its mean is 0.204 in our sample, indicating that 20.4% of observations are in retail and service industries (Table 1 Panel B).

4.2 Regression analyses

4.2.1 Determinants of changes in managers' GAI discussion intensity

To examine how the four firm and industry characteristics of interest are associated with changes in managers' GAI discussion intensity in conference calls surrounding the release of

¹⁹ The data of major customers used in the literature to measure customer-base concentration (e.g., Patatoukas 2012; Crawford et al. 2020) is not helpful for our research purpose because we need to include small individual customers.

ChatGPT, we run a regression of $\Delta GAI\%$ on the measures of these characteristics, separately and collectively:

$$\Delta GAI\%_{i,t} = \alpha + \beta X_{i,t-1} + \lambda Controls_{i,t-1} + \varepsilon_{i,t}, \quad (1)$$

where the independent variable of interest, $X_{i,t-1}$, is *Patents*, *Trade Secret*, measures of R&D expenditure (*R&D* and *Miss R&D*), *CS Threat*, *AIIE*, or *Customer*. *Controls* refers to control variables. We first control for the fundamental firm and industry characteristics we examine in Section 3, including *Firm Size*, *Market to Book*, *Firm Age*, and *High Tech Industry*—an indicator for the high-tech industries. As *AIIE* and *Customer* are measured at the industry level, we do not include industry fixed effects. Studies find that firms with greater investor interest are more likely to host conference calls (Tasker 1998; Frankel et al. 1999). To control for investor information demand, we include *Analyst Coverage*, measured by the number of financial analysts following a firm, and *Institutional Shareholding*, measured by the percentage of shares outstanding owned by institutional investors. As studies find that firms with less return volatility tend to provide more voluntary disclosures (Li 2010), we also control for *Abnormal Return Volatility*, measured by the standard deviation of market-adjusted return. We winsorize continuous variables at the 1 percent and 99 percent levels and cluster the standard errors at the industry (three-digit SIC) level.

Table 3 Panel A presents the estimation results. To facilitate interpretation and comparison across regressions, in this and all subsequent analyses, we normalize $\Delta GAI\%$, *Patents*, *R&D*, *CS Threat*, and *AIIE* by subtracting the sample mean and then dividing by the standard deviation. In Column (1), we include only the control variables to estimate a benchmark model as well as to examine how the fundamental firm and industry characteristics examined in Sections 3.4.2 and 3.4.3 are associated with $\Delta GAI\%$ in a multivariate regression. Consistent with the univariate evidence in Sections 3.4.2 and 3.4.3, *Size*, *Market to Book*, and *High Tech Industry* load positively

and significantly, with a coefficient of 0.04 ($t\text{-stat} = 2.74$), 0.01 ($t\text{-stat} = 2.64$), and 0.29 ($t\text{-stat} = 1.93$), respectively, and *Firm Age* loads negatively and significantly, with a coefficient of -0.04 ($t\text{-stat} = -2.24$). These coefficients are also economically significant. A one-standard deviation increase in *Size* and *Market to Book* corresponds to a 0.08 and 0.14 standard deviation increase in $\Delta GAI\%$, while a one-standard deviation increase in *Firm Age* corresponds to a 0.80 standard deviation decrease in $\Delta GAI\%$. Compared with non-high-tech industries, a high-tech industry firm has a 0.29 standard deviation increase in $\Delta GAI\%$. These results suggest that managers of larger, younger, growth, and high-tech firms are more likely to increase GAI discussions.

In Columns (2)–(7), the six measures of innovation, cybersecurity threats, labor exposure to AI, and customer operations are included individually, and in Column (8) they are included simultaneously. We find that all measures, except the measures of trade secrets and R&D expenditures, load positively and significantly, both individually and collectively. When all variables are included (Column (8)), *Patents* has a coefficient of 0.06 ($t\text{-stat} = 2.30$), *CS Threat* of 0.05 ($t\text{-stat} = 2.25$), *AIIE* of 0.07 ($t\text{-stat} = 2.87$), and *Customer* of 0.51 ($t\text{-stat} = 3.94$). These coefficients are also economically significant. A one-standard deviation increase of *Patents* corresponds to a 0.06 standard deviation increase in $\Delta GAI\%$. As the mean and standard deviation of $\Delta GAI\%$ are 0.306 and 1.609, respectively, this effect accounts for a 32% increase ($= 0.06 \times 1.609 / 0.306$) of the sample mean. A similar calculation suggests that a one-standard deviation increase in *CS Threat* (*AIIE*) corresponds to an increase of $\Delta GAI\%$ that accounts for 26% (37%) of its sample mean. As *Customer* is an indicator variable, its coefficient 0.51 indicates that $\Delta GAI\%$ is larger by 0.51 standard deviation in industries with high customer operations (retail and services industries) than others, which accounts for 268% ($= 0.51 \times 1.609 / 0.306$) of the sample

mean of $\Delta GAI\%$.²⁰ In addition, *Size*, *Market to Book*, *Firm Age*, and *High Tech Industry* continue to load significantly and consistently across the specifications. Overall, the results in Table 3 Panel A indicate that the increase in managers' GAI-related discussions in conference calls after the release of ChatGPT is greater for larger, younger, growth, and high-tech firms, as well as firms with more patent filings, greater cybersecurity threats, higher labor exposure to AI, and more customer operations, suggesting that these firms are perceived by managers to be more impacted by GAI.

The different findings for patents, trade secrets, and R&D expenditures suggest that the association between innovation intensity and changes of managers' GAI discussions is subtle. While patents and trade secrets are alternative mechanisms to protect innovation output, the latter are subject to weaker protection (Hall et al. 2014). A trade secret may be legally appropriated by a firm that has independently discovered or reverse engineered it, whereas a patent may not. Hence, firms with more patents may view GAI as presenting greater opportunities to foster innovation without fearing appropriation by competitors.²¹ R&D expenditure, on the other hand, captures innovation inputs whose outcome is highly uncertain, and thus how firms with higher R&D expenditures are affected by GAI is unclear.

For comparison, we also investigate changes of analysts' GAI discussions using $\Delta GAI\%$ -*Analyst* as the dependent variable and report the results in Table 3 Panel B. We find that financial analysts increase GAI discussions more for firms with greater cybersecurity threats, higher labor exposure to AI, and more customer operations, like the evidence for managers. However, unlike

²⁰ The coefficient of *Customer* is larger than that of *High Tech Industry* and the difference is significant based on an untabulated test. This suggests that in managers' views, retail and services industries are more affected by GAI than high-tech industries.

²¹ Another possible reason is that trade secrets are much broader than unpatented innovations, including confidential financial data, customer lists, business plans, and so on (Contigiani et al. 2018; Klasa et al. 2018), whose relations with GAI could be complex.

the evidence for managers, the association between $\Delta GAI\%-Analyst$ and *Patents* is insignificant. As analysts' questions follow managers' presentations, those discussions reflect their perceptions and demand for additional information.²² The insignificant result for *Patents* suggests that analysts do not demand more information of GAI for firms with more patents after managers' discussions. Overall, the evidence suggests that analysts' perceptions are largely consistent with those of the managers.

4.2.2 *Changes of managers' GAI discussions with different sentiment*

While the analysis in Table 3 speaks to which types of firms' managers perceive their firms to be more affected by GAI, the directions of the impacts are unclear. To address this question, we examine how the firm and industry characteristics are associated with changes of managers' GAI discussions with positive, negative, and neutral tones. We re-estimate Equation (1) using $\Delta GAI\%-Pos$, $\Delta GAI\%-Neg$, and $\Delta GAI\%-Neu$ as the dependent variables. The findings are reported in Table 4 Panel A. Again, we normalize the dependent variables and *Patents*, *R&D*, *CS Threat*, and *AIIE*. This makes the comparison cross regressions not confounded by the relative frequencies of GAI discussions of different tones.²³

The coefficients of *Patents* and *CS Threat* are positive and significant for $\Delta GAI\%-Pos$ as the dependent variable (Column (1)), while they are insignificant for $\Delta GAI\%-Neg$ (Column (2)) and $\Delta GAI\%-Neu$ (Column (3)). Furthermore, the coefficients in the Column (1) are significantly more positive than the corresponding ones in Column (2). These results indicate that, for firms with more patent filings or cybersecurity threats, managers increase positive-tone GAI discussions more

²² For this reason, in an unreported robustness test, we include changes of managers' GAI discussions in the presentation section as an additional control to account for the possibility that the analysts' questions are prompted by the managers' discussions in the presentation section. We find similar results. We do not include this control in our tabulated analyses to avoid the problem of a bad control (Angrist and Pischke 2009).

²³ One issue with this approach is that, while most of managers' GAI discussions have positive tones, the three types of discussions are treated as equally important. To address this issue, we use unnormalized dependent variables and report the results in Table IA-2 of in the internet appendix. The findings are similar.

but not negative- or neutral-tone discussions, suggesting that managers tend to view GAI's impact on these firms to be positive. The coefficients of the measures of trade secrets and R&D expenditures are insignificant in all regressions. Thus, trade secrets and R&D expenditures are unassociated not only with overall changes of GAI discussion intensity but also those of specific tones.

The coefficients on *AIIE* and *Customer* are positive and significant in all columns, suggesting that managers of firms with greater labor exposure to AI and more customer operations are more likely to increase GAI discussions of all tones. However, the coefficients are significantly more positive for positive-tone discussions (Column (1)) than for negative-tone ones (Column (2)), which suggests that, for firms with higher *AIIE* and *Customer*, managers' perceptions of GAI's impact are more likely to be positive.

These results suggest that managers of firms with more patents and firms facing greater cybersecurity threats view GAI positively, while the views of managers of firms with greater labor exposure to AI and firms with more customer operations appear to be more varied, though they are more likely to be positive than negative. We again report the results based on analysts' GAI discussions in Panel B for comparison. The results are consistent with those for managers in Panel A, except that *Patents* is insignificantly associated with changes in positive-tone analyst discussions. This comparison resembles that in Table 3.

To examine the three types of discussions together, we construct a broad measure of positive-tone GAI discussions by combining *GAI%-Pos*, *GAI%-Neg*, and *GAI%-Neu* using the weighting scheme 1, -1, and 0. Specifically, for each conference call, we define *GAI%-Pos-Broad* as *GAI%-Pos* minus *GAI%-Neg*. We then calculate the change of this composite measure for each firm after the launch of ChatGPT, denoted as $\Delta GAI\%-Pos-Broad$. We use this variable as the dependent

variable and estimate Equation (1). As reported Table IA-3 of the internet appendix, we continue to find that *Patents*, *CS Threat*, *AIIE*, and *Customer* are positively associated with changes of the broadly defined positive-tone discussions.

4.2.3 *Changes of managers' GAI discussions with and without initiatives*

When discussing GAI, managers may disclose specific initiatives or comment broadly without initiatives. For example, Balakrishnan et al. (2023) find that, when managers discuss racial diversity issues in conference calls, some mention specific actions, such as appointing Black directors, while others offer general remarks without presenting plans. Thus, we further investigate how the firm and industry characteristics we study are associated with the changes of managers' GAI discussions with and without initiatives. Using $\Delta GAI\%-Initiative$ and $\Delta GAI\%-NonInitiative$ as dependent variables, we re-estimate Equation (1) and report the results in Table 5 Panel A. As in Tables 3 and 4, we normalize the dependent variables and *Patents*, *R&D*, *CS Threat*, and *AIIE*. The coefficients on *Patents*, *CS Threat*, and *Customer* are significantly positive when $\Delta GAI\%-Initiative$ is the dependent variable (Column (1)) but insignificant when $\Delta GAI\%-NonInitiative$ is the dependent variable (Column (2)). Moreover, the difference between each pair of corresponding coefficients is significant. These results suggest that, for firms with more patent filings, greater cybersecurity threats, and more customer operations, managers increase GAI discussions with initiatives more after the release of ChatGPT, while there are no such associations for discussions without initiatives.

The coefficients of *AIIE* are significantly positive in both columns, with the one in Column (1) significantly exceeding that in Column (2). This result suggests that managers of firms with greater labor exposure to AI increase discussions of both types more after the release of ChatGPT but the increase is greater for initiative-related discussions. As in Tables 3 and 4, trade secrets and

R&D expenditures are insignificantly associated with changes in both types of discussion. The results for analyst discussions reported in Panel B are consistent with those for managers, except that *Patents* relates insignificantly to changes in analyst initiative discussions.

4.3 AI washing?

One concern may be that managers' GAI discussions in conference calls reflect AI washing. We provide three pieces of evidence that alleviate these concerns. First, we categorize the discussions along four non-mutually exclusive dimensions using ChatGPT-4 (guided by and verified with manual reading). Eight-seven percent of GAI initiative discussions entail actions that are already taken; 74% of discussions are specific; 38% entail partnership with a GAI vendor; and 68% are with sufficient certainty. Considering the significant legal liabilities for managers to mislead investors in conference calls, these features are inconsistent with AI washing. Second, we ask ChatGPT-4 to read all initiative-related discussions to determine whether they are likely AI washing.²⁴ Only nine out of 312 calls (2.9%) are identified as AI washing. Twenty-one calls (6.7%) are identified as "not sure," and the remaining 90.4% are identified as not AI washing.²⁵

Finally, we posit that because AI talent is crucial to firms' AI strategy, if firms truly undertake GAI initiatives, they are likely to hire more AI talent. To test this conjecture, we compare AI-related job postings of GAI discussing and non-discussing firms during the year after ChatGPT's launch using the Revelio Labs database that provides workforce data. We identify AI-related jobs based on manual coding of the job classifications. We report the results in Table 6. Among firms

²⁴ We use the following prompt: *Does the manager's discussion of real initiatives or plans about generative AI below in a company's conference call reflect AI washing? AI washing refers to the practice of companies making false, misleading, or exaggerated statements about whether and how they're using or developing artificial intelligence systems. Note that managers face legal liabilities if making statements that are materially misleading or inaccurate ex-post in a conference call. Answer with 1 for 'Yes', 0 for 'No', and -1 for 'Not sure'. Text:{}*.

²⁵ In an untabulated analysis, we repeat the main analysis by excluding the nine calls identified as AI washing by ChatGPT, and our inferences remain the same.

discussing GAI, an average of 365 AI-related jobs are posted, significantly exceeding the 103 AI-related jobs posted, on average, by firms that do not discuss GAI. Moreover, the percentage of AI postings relative to total postings is significantly higher for firms that discuss GAI compared to those that do not. The inferences are similar when we compare firms with initiative-related GAI discussions and those without GAI discussions, with the differences slightly larger. Taken together, our evidence mitigates the concern that managers' GAI discussions may primarily reflect AI washing. However, we acknowledge that we cannot rule out this concern and leave the further exploration of this issue for future research.

5 Market reactions to changes in managers' GAI discussions

In this section, we examine market reactions to managers' GAI discussions in conference calls to assess whether and how the discussions are informative to investors. We measure market reactions with the absolute value of abnormal returns ($|Call\ CAR|$) and abnormal trading volume ($Ab. Volume$) over the two-day window $[0, +1]$, where day 0 is the conference call dates. We regress $|Call\ CAR|$ and $Ab. Volume$ on $\Delta GAI\%$ and a set of control variables measured at the closest fiscal quarter (e.g., Bushee et al. 2011; Frankel et al. 1999) and report the results in Table 7.²⁶ We find that changes in managers' GAI discussions are significantly and positively associated with both $|Call\ CAR|$ and $Ab. Volume$ (Columns (1) and (2) in Panel A). Specifically, the coefficient for $\Delta GAI\%$ is 0.03 for abnormal returns and 0.05 for abnormal trading volume, both significant at the 1% level. We find similar results for $\Delta GAI\%-Pos-Broad$ (Columns (3) and (4)), the change of the broad measure of positive-tone discussions defined in Section 4.2.2, and $\Delta GAI\%-Initiative$ (Columns (5) and (6)). These results suggest that changes of managers' GAI discussions (intensity and features) do inform investors.

²⁶ The control variables are quarter-end market value, market-to-book ratio, return volatility, absolute earnings surprises, absolute call sentiment, and quarterly return on assets.

Next, we examine whether the market reactions vary with the level of institutional ownership, to assess whether institutional or retail investors drive these responses. We categorize firms into two groups: high and low institutional shareholding, based on the highest and lowest sample tercile. The results presented in Table 7 Panels B and C show that market reactions are primarily concentrated among firms with high institutional shareholding. While both groups exhibit significant reactions to $\Delta GAI\%$ and $\Delta GAI\%-Initiative$, only the high institutional shareholding group shows significant reactions to $\Delta GAI\%-Pos-Broad$. The coefficients for firms with high institutional ownership are significantly larger than those for the low institutional ownership group, except for $\Delta GAI\%-NonInitiative$. Overall, these findings suggest that institutional investors primarily drive the market responses.

Finally, to explore the possibility that initiative-related discussions with distinct features may have varying information content, we explore whether the market reacts differently to changes of initiative-related GAI discussions with different features discussed in the first point of Section 4.3 (e.g., specificity and whether the action has occurred), and report the results in Table 7 Panel D. For initiatives that have occurred versus those that have not, the reactions reflected in $|Call\ CAR|$ are both significant and have similar magnitudes. However, the reaction based on $Ab. Volume$ is significant only for the former. The market reacts significantly to discussions of specific initiatives and those with sufficient certainty, based on both measures, whereas the reactions are insignificant for vague initiatives and those without sufficient certainty. The market reactions to initiatives involving a partnership with a GAI vendor do not differ significantly from those to initiatives not mentioning a partnership. Overall, we find that initiative-related GAI discussions have greater information content when they are more specific and more certain.

6 Additional analyses

6.1 Robustness tests

In all regression analyses above, we use the *change* of GAI discussions or those with certain content features (sentiment and initiative-relatedness) as the dependent variable to examine managers' *responses* to the release of ChatGPT. To understand how managers and analysts discuss GAI cross-sectionally in the post-ChatGPT period, we use the *level* of discussions (e.g., *GAI%* and *GAI%-Pos*) as the dependent variable and repeat the analyses in Tables 3–5. As reported in Table 8 Panel A, the results are generally consistent with those in Tables 3–5. Thus, all our conclusions regarding changes in managers' GAI discussions hold for the level of discussions in the post-ChatGPT period. This is expected because managers rarely discuss GAI before ChatGPT. Next we conduct a robustness test for the analyses in Tables 3–5 by measuring GAI discussions based on the number of words (as opposed to sentences) relative to the total number of words in managers' disclosures in a call. The results reported in Table 8 Panel B are also consistent with those in Tables 3–5.

6.2 Within-industry analyses

We do not include industry fixed effects in Equation (1) because *AIIE* and *Customer* are measured at the industry level. Thus, within- and between-industry variations both drive our findings regarding innovation intensity and cybersecurity threats. To explore whether within-industry variations of these three factors have similar effects, we incorporate industry (three-digit SIC) fixed effects in Equation (1). The findings are detailed in Table 9. For managers' GAI discussions in Panel A, *Patents* and *CS Threat* continue to have significant and positive coefficients for $\Delta GAI\%$, $\Delta GAI\%-Pos$, and $\Delta GAI\%-Initiative$. For analysts' GAI discussions in Panel B, *CS Threat* continues to have a significant and positive coefficient for $\Delta GAI\%-Analyst$, $\Delta GAI\%-Pos-Analyst$, and $\Delta GAI\%-Initiative-Analyst$. These results suggest that within-industry

variations of innovation (*Patents*) and cybersecurity threats also contribute to the managers' results above, and within-industry variations of cybersecurity threats also contribute to the analysts' results above.

6.3 Market reaction to launch of ChatGPT

Finally, we explore whether managers' perceptions of GAI's impact are consistent with those of investors by examining market reaction to the launch of ChatGPT on November 30, 2022. We measure the market reaction by the cumulative abnormal return over the two-day window $[0, +1]$, where day 0 is the ChatGPT launch date (*GPT Launch CAR*) and regress the market reaction on the measures of firm and industry characteristics of interest. The results are reported in Table 10. In Columns (1)–(6), we include one measure at a time, and in Column (7), we include all measures. Consistent with our findings from managers' GAI discussions, *Patents* and *CS Threat* are positively and significantly associated with *GPT Launch CAR*, with coefficients of 0.19 ($t\text{-stat} = 3.15$) and 3.45 ($t\text{-stat} = 2.25$), respectively. This result suggests that investors perceive GAI to benefit firms with more patents and greater cybersecurity threats.

However, unlike our findings based on managers' GAI discussions, *Trade Secret* and *R&D* are negatively and significantly associated with *GPT Launch CAR*, with coefficients of -0.83 ($t\text{-stat} = -2.25$) and -4.52 ($t\text{-stat} = -10.09$), respectively, while both *AIIE* and *Customer* do not load significantly.²⁷ Thus, it seems that investors perceive GAI to hurt firms with higher R&D expenditures and those with more trade secrets, while their perception of GAI's impact does not vary with labor exposure to AI or customer operations. As we discuss in Section 4, GAI's impacts on firms with different levels of innovation intensity, labor exposure to AI, and customer operations are unclear. The results in Table 10 suggest that, along these dimensions, managers'

²⁷ When all variables of interest are included simultaneously in Column (7), the coefficient of *AIIE* becomes significantly positive, which is consistent with the finding based on managers' GAI discussions.

perceptions seem to be more positive than those of investors. These inconsistencies are understandable because ChatGPT is a breakthrough technology and its impacts on the business world are uncertain. Thus, investors' perceptions, reflected in price changes within several days, could be inconsistent with those of managers. In this case, examining the perceptions and responses of various economic agents (investors, managers, analysts, and others) can better depict GAI's impacts.

GAI announcement returns may also be confounded by other important market-wide events occurring on November 30, 2022. For example, on that same day, Federal Reserve Chair Jerome Powell delivered a speech indicating the possibility of slowing the pace of interest rate hikes, and the Commerce Department released a higher-than-expected GDP growth rate. While we adjust our return measure for the market return, the impacts of confounding events could vary with the firm and industry characteristics we study, contaminating our return analysis. Note that these confounding events do not contaminate our analyses of conference calls because we specifically examine GAI discussions. Moreover, calls may release other new information that we cannot control for, so our findings should be interpreted with appropriate caution.

7 Conclusion

We provide early evidence of firms' responses to GAI through managerial discussions of GAI in conference calls after the release of ChatGPT in November 2022. We find that, following the release, these managerial discussions increase substantially, and most discussions have positive tones and contain action initiatives. The increase is more pronounced for larger firms, growth firms, younger firms, and high-tech industries. The largest increase is observed in the Fama-French-12 industry of business equipment-computers, software, and electronic equipment, while the utilities, chemicals and allied products, and oil, gas, and coal extraction industries experience

the smallest increases. We further examine how the change in managers' GAI discussions is associated with several important firm and industry characteristics related to the major economic channels through which GAI is generally believed to impact firms, including innovation, cybersecurity, labor, and customers. We find that managers of firms with greater innovation intensity (reflected in patent filings), cybersecurity threats, labor exposure to AI, and customer operations are more likely to increase GAI discussions.

Our analysis of discussion sentiment suggests that managers of firms with more patent filings and those with greater cybersecurity threats hold positive views of GAI. While managers of firms with greater labor exposure to AI and those with more customer operations hold mixed views, their perspectives are more likely to be positive than negative. Additional content analysis suggests that managers of firms with more patent filings, greater cybersecurity threats, and more customer operations increase initiative-related discussions more but not discussions without initiatives. While managers of firms with greater labor exposure to AI increase discussions of both types more, they are more likely to increase initiative-related ones. We also find that investors, especially institutional investors, find changes of managers' GAI discussions in conference calls following the launch of ChatGPT informative.

We find similar results for changes in analysts' GAI discussions in conference calls, except that analysts' responses generally do not vary with firms' patent filings. Exploring market reactions to the launch of ChatGPT, we find that investors' perceptions are consistent with managers in some but not all dimensions. Overall, our study illuminates the heterogeneous corporate responses to GAI.

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Appendix A

Sample Selection

	Number of Firms	Number of Calls
Pull out the latest firm-years with positive total assets from Compustat between Nov 29, 2021, and Nov 30, 2022 (ChatGPT launch date)	8,065	-
Keep firms with necessary firm-level variables	2,869	-
Merge conference call data from Nov 30, 2022, to May 27, 2023	2,280	6,772
Keep calls without missing necessary call-level variables	2,025	6,192

Appendix B

GAI-Related Keywords

This appendix provides a list of GAI-related keywords that we use to identify GAI discussions in conference calls.

Vendor	Product	Technology
OpenAI	GPT	Large language model
Lightricks	GPT3	Prompt Learning
Stability AI	GPT3.5	ControlNet
Google AI	GPT4	Text To Image
Google Deepmind	ChatGPT	GAN
Cohere	MakerSuite	Transformer
Hugging Face	PaLM API	LLM
Bard	Rerank	Generative AI
Whisper	AlphaFold	VAE
Midjourney	Stable Diffusion	Generative adversarial network
Claude	Copilot	Natural language generation
Glean	Power BI	Diffusion model
Jasper	DALL-E	Variational autoencoder
Anthropic		Generative pre-trained transformer

Appendix C

Examples of Managers' GAI Discussions

This appendix provides examples of managers' GAI discussions during conference calls. Panel A presents examples based on the discussion sentiment (positive, negative, or neutral). Panel B presents examples based on the discussion content (initiative vs. non-initiative). Panel C presents examples based on the discussion topics that pertain to the four key dimensions of GAI's impact examined in this paper.

Panel A: Discussion Sentiment

Sentiment	Company	Call Date	Manager Discussion
Positive	Software Aktiengesellschaft	April 27, 2023	"Another great innovation and achievement is the integration of ChatGPT. In January, we became the first independent iPaaS platform provider to introduce a ChatGPT API connector enabling seamless interaction with this innovative language model. Innovation like this continues to increase the impact we can deliver for customers. For example, we recently secured an outstanding win with Lufthansa Technik. Our data integration solution web methods has enabled them to reduce manual processes on 45,000 transactions per day from 27% to 8%. This has helped not only lower costs and increase operational performance but also enhance the technological competitive advantage."
Negative	Sumitomo Pharma Co., Ltd	April 28, 2023	"You also ask how we are using ChatGPT. We have a digital team in the U.S., and we are using the same technology. However, I have heard that even with ChatGPT, there are some erroneous results. So I don't think we can trust the system fully, but I am aware of its use as 1 of the tools used at work."
Neutral	Kakao Corp.	February 9, 2023	"On top of reinforcing our core business, we are exploring new growth opportunities through our focus on AI and Healthcare. We see hyperscale AI model, including ChatGPT, which has recently been the buzz, will be both an opportunity and a threat to Kakao. I say this because services developed based on AI are not differentiated by innovative ideas, but by the size of the model which is determined by the amount of investment you put in. And because the quality is dictated by investment, global companies with their large sum of capital and technologies have an absolute upper hand in this battle."

Panel B: Discussion Content

Type	Company	Call Date	Manager Discussion
Initiative	Farfetch Limited	May 18, 2023	"One of the areas we are most excited about is the recent developments that large language models are bringing to the field of AI. We're already in a leading position with respect to AI intellectual industry with significant in-house data science, AI, and machine learning teams, and we have been active in this space for many years now. Our long-standing partnership with Microsoft has opened up the opportunity to access the most advanced version of ChatGPT, and our tech teams have been developing several concrete applications of ChatGPT for the luxury space. I believe this could be a significant development for Farfetch."
Non-Initiative	Nepa AB	May 16, 2023	"We will always have a need, this really clever human touch on what we deliver. And even if ChatGPT is getting better and better by the day, yes. So far, my colleagues are clever, more clever than ChatGPT."

Appendix C (Continued)
Examples of Managers' GAI Discussions

Panel C: Discussion Topic			
Topic	Company	Call Date	Manager Discussion
Innovation	Dialogue Health Technologies Inc	May 10, 2023	“There's a lot happening in AI, large language models and all sorts of technologies that are coming up. The way we think about it is patient-facing and nonpatient facing, right? So in nonpatient-facing opportunities, we're really taking advantage of ChatGPT and others. So for example, our developers are using ChatGPT to help with code the use that GBT to do unit tests on code and to find solutions and answers to some of their problems. So that's been used quite a bit. We had a internal hackathon a couple of weeks ago. A lot of the ideas were around ChatGPT.”
Cybersecurity	Palo Alto Networks, Inc	March 7, 2023	“And what it does is it causes havoc in a company over here and everybody has to go upgrade the entire technical stack. So I don't think cybersecurity is going away. I think cybersecurity is going to be here to stay. I think there's a lot of technical debt we have not paid in the cybersecurity space. Cybersecurity -- cyber risk is now the #1 enterprise risk for companies, one that they can try and manage, excluding pandemics, et cetera. So it's a growth market. So in that market, with the huge technological change that is going on now with the advent of AI, the people are using AI right malware. We're testing in our labs. We're trying to figure out if we can write malware and go protect ourselves against malware written by ChatGPT. So try doing that, right?”
Labor	Coursera, Inc.	March 9, 2023	“Now the big, what about, is generative AI. I've been absolutely obsessed with ChatGPT probably most of you have as well. My view is, this is a fundamental game changer. Like I don't even know what society is going to be like in 18, 36 months. When every major software platform builds that ChatGPT and other generative AI technologies into their software, it's going to show up before people even know what to do with it. So I don't know if generative AI is going to take a lot of these knowledge jobs and make them more vulnerable as well. But in any case, the rate of change driven by technology and globalization is creating major change in the way that people do jobs, the way that businesses run and the skills and the knowledge and the credentials that people need to effectively add value to these companies. So as Bain said, in 2022, this is more true now than ever. The big challenge for all of us is trying to figure out how to transition workers from declining occupation to jobs of the future, especially in a world where a job of the future that we thought like 6 months ago might be now declining occupation. I mean it is changing so fast, it's mind-blowing.”
Customer	SK Telecom Co., Ltd	February 8, 2023	“And moving on to ChatGPT, we already have the GPT-3 in A. service. So we have accumulated a large amount of customer data. So by using this data, we are advancing the ChatGPT-type technology. So there is fierce competition with big tech companies globally. So in order to ensure rapid upgrades and improvement of A., we are considering various options and solutions, including those from the overseas. And the direction of advancing our technology will be in line with the kinds of services that we want to offer to our customers. So the adoption of long-term memory technology and the multimodal technology are very much in line with this direction. In doing so, we will be able to offer differentiated services to our customers based on a differentiated conversation and AI technology.”

Appendix D

Variable Definitions

Variable	Definition
GAI Discussion Variables	
<i>GAI%</i>	Managers' GAI discussion intensity, measured as the percentage of GAI-related sentences in their presentation and answers to analysts' questions in a conference call, multiplied by 100. We identify GAI-related sentences following a two-step procedure. We first identify sentences containing keywords listed in Appendix A, we then manually read sentences centering around the keyword sentences to identify GAI contextual sentences. GAI-related sentences include both the keyword sentences and the contextual sentences.
$\Delta GAI\%$	<i>GAI%</i> in post-ChatGPT period minus the average <i>GAI%</i> for the same firm in the one-year pre-ChatGPT period.
<i>GAI%-Pos</i>	Equals <i>GAI%</i> if the net sentiment of GAI discussions is positive, and zero otherwise.
$\Delta GAI\%-Pos$	<i>GAI%-Pos</i> in post-ChatGPT period minus the average <i>GAI%-Pos</i> for the same firm in the one-year of pre-ChatGPT period.
<i>GAI%-Neg</i>	Equals <i>GAI%</i> if the net sentiment of GAI discussions is negative, and zero otherwise.
$\Delta GAI\%-Neg$	<i>GAI%-Neg</i> in post-ChatGPT period minus the average <i>GAI%-Neg</i> for the same firm in the one-year pre-ChatGPT period.
<i>GAI%-Neu</i>	Equals <i>GAI%</i> if the net sentiment of GAI discussions is neutral, and zero otherwise.
$\Delta GAI\%-Neu$	<i>GAI%-Neu</i> in post-ChatGPT period minus the average <i>GAI%-Neu</i> for the same firm in the one-year pre-ChatGPT period.
<i>GAI%-Initiative</i>	Equals <i>GAI%</i> if the GAI discussion contains GAI-related initiatives, and zero otherwise. We use ChatGPT-4 to identify whether a corpus of text contains initiative of adopting GAI. Our prompt is as follows: Does the text involve real initiatives or plans in generative AI? Answer with 1 for "Yes", 0 for "No". Text: { }.
$\Delta GAI\%-Initiative$	<i>GAI%-Initiative</i> in post-ChatGPT period minus the average <i>GAI%-Initiative</i> for the same firm in the one-year pre-ChatGPT period.
<i>GAI%-NonInitiative</i>	Equals <i>GAI%</i> if the GAI discussion exists and does not contain GAI-related initiatives, and zero otherwise. We use ChatGPT-4 to identify whether a corpus of text contains initiative of adopting GAI. Our prompt is as follows: Does the text involve real initiatives or plans in generative AI? Answer with 1 for "Yes", 0 for "No". Text: { }.
$\Delta GAI\%-NonInitiative$	<i>GAI%-NonInitiative</i> in post-ChatGPT period minus the average <i>GAI%-NonInitiative</i> for the same firm in the one-year pre-ChatGPT period.
<i>GAI%-Analyst</i>	Analysts' GAI discussion intensity, measured as the percentage of GAI-related sentences in their questions in a conference call, multiplied by 100. We identify GAI-related sentences following a two-step procedure. We first identify sentences containing keywords listed in Appendix A, we then manually read sentences centering around the keyword sentences to identify GAI contextual sentences. GAI-related sentences include both the keyword sentences and the contextual sentences.
$\Delta GAI\%-Analyst$	<i>GAI%-Analyst</i> in post-ChatGPT period minus the average <i>GAI%-Analyst</i> for the same firm in the one-year of pre-ChatGPT period.

Appendix D (Continued)

Variable Definitions

Variable	Definition
<i>GAI%-Pos-Analyst</i>	Equals <i>GAI%-Analyst</i> if the net sentiment of analysts' GAI discussions is positive, and zero otherwise.
$\Delta GAI\%-Pos-Analyst$	<i>GAI%-Pos-Analyst</i> in post-ChatGPT period minus the average <i>GAI%-Pos-Analyst</i> for the same firm in the one-year pre-ChatGPT period.
<i>GAI%-Neg-Analyst</i>	Equals <i>GAI%-Analyst</i> if the net sentiment of analysts' GAI discussions is negative, and zero otherwise.
$\Delta GAI\%-Neg-Analyst$	<i>GAI%-Neg-Analyst</i> in post-ChatGPT period minus the average <i>GAI%-Neg-Analyst</i> for the same firm in the one-year pre-ChatGPT period.
<i>GAI%-Neu-Analyst</i>	Equals <i>GAI%-Analyst</i> if the net sentiment of analysts' GAI discussions is neutral, and zero otherwise.
$\Delta GAI\%-Neu-Analyst$	<i>GAI%-Neu-Analyst</i> in post-ChatGPT period minus the average <i>GAI%-Neu-Analyst</i> for the same firm in the one-year pre-ChatGPT period.
<i>GAI%-Initiative-Analyst</i>	Equals <i>GAI%-Analyst</i> if analysts' GAI discussions contain GAI-related initiatives, and zero otherwise. We use ChatGPT-4 to identify whether a corpus of text contains initiative of adopting GAI. Our prompt is as follows: Does the text involve real initiatives or plans in generative AI? Answer with 1 for "Yes", 0 for "No". Text: {}.
$\Delta GAI\%-Initiative-Analyst$	<i>GAI%-Initiative-Analyst</i> in post-ChatGPT period minus the average <i>GAI%-Initiative-Analyst</i> for the same firm in the one-year pre-ChatGPT period.
<i>GAI%-NonInitiative-Analyst</i>	Equals <i>GAI%-Analyst</i> if analysts' GAI discussion exists and does not contain GAI-related initiatives, and zero otherwise. We use ChatGPT-4 to identify whether a corpus of text contains initiative of adopting GAI. Our prompt is as follows: Does the text involve real initiatives or plans in generative AI? Answer with 1 for "Yes", 0 for "No". Text: {}.
$\Delta GAI\%-NonInitiative-Analyst$	<i>GAI%-NonInitiative-Analyst</i> in post-ChatGPT period minus the average <i>GAI%-NonInitiative-Analyst</i> for the same firm in the one-year pre-ChatGPT period.
$\Delta GAI\%-Pos-Broad$	Equals $\Delta GAI\%-Pos$ minus $\Delta GAI\%-Neg$.
<i>GAI%-Initiative (Already Happened)</i>	Equals <i>GAI%-Initiative</i> if the GAI discussion contains GAI-related initiatives that have already happened, and zero otherwise. We use ChatGPT-4 to identify these initiatives. Our prompt is as follows: Does the text discuss a real generative AI related action that already happened? Answer with 1 for "Yes", 0 for "No". Text: {}.
$\Delta GAI\%-Initiative (Already Happened)$	<i>GAI%-Initiative (Already Happened)</i> in post-ChatGPT period minus the average <i>GAI%-Initiative (Already Happened)</i> for the same firm in the one-year pre-ChatGPT period.
<i>GAI%-Initiative (Specific)</i>	Equals <i>GAI%-Initiative</i> if the GAI discussion contains specific GAI-related initiatives, and zero otherwise. We use ChatGPT-4 to identify these initiatives. Our prompt is as follows: Is the text that discusses real initiatives or plans in generative AI specific? Specificity includes mentioning a product, an application, a number, a function, or an example. Answer with 1 for "Yes", 0 for "No". Text: {}.
$\Delta GAI\%-Initiative (Specific)$	<i>GAI%-Initiative (Specific)</i> in post-ChatGPT period minus the average <i>GAI%-Initiative (Specific)</i> for the same firm in the one-year pre-ChatGPT period.
<i>GAI%-Initiative (Partner)</i>	Equals <i>GAI%-Initiative</i> if the GAI discussion contains GAI-related initiatives that relate to a specific business partner, and zero otherwise. We use ChatGPT-4 to identify these initiatives. Our prompt is as follows: Is the text that discusses real initiatives or plans in generative AI specific? Specificity includes mentioning a product, an application, a number, a function, or an example. Answer with 1 for "Yes", 0 for "No". Text: {}.
$\Delta GAI\%-Initiative (Partner)$	<i>GAI%-Initiative (Partner)</i> in post-ChatGPT period minus the average <i>GAI%-Initiative (Partner)</i> for the same firm in the one-year pre-ChatGPT period.

Appendix D (Continued)

Variable Definitions

<i>GAI%-Initiative (Certain)</i>	Equals <i>GAI%-Initiative</i> if the GAI discussion of GAI-related initiatives is sufficiently certain, and zero otherwise. We use ChatGPT-4 to identify these initiatives. Our prompt is as follows: Does the text that involves real initiatives or plans in generative AI discuss the initiatives or plans with sufficient certainty? Answer with 1 for “Yes”, 0 for “No”. Text: {}.
<i>ΔGAI%-Initiative (Certain)</i>	<i>GAI%-Initiative (Certain)</i> in post-ChatGPT period minus the average <i>GAI%-Initiative (Certain)</i> for the same firm in the one-year pre-ChatGPT period.
<i>GAI%-Initiative (Not Already Happened)</i>	Equals <i>GAI%-Initiative</i> if the GAI discussion does not contain GAI-related initiatives that have already happened, and zero otherwise. We use ChatGPT-4 to identify these initiatives. Our prompt is as follows: Does the text discuss a real generative AI related action that already happened? Answer with 1 for “Yes”, 0 for “No”. Text: {}.
<i>ΔGAI%-Initiative (Not Already Happened)</i>	<i>GAI%-Initiative (Not Already Happened)</i> in post-ChatGPT period minus the average <i>GAI%-Initiative (Not Already Happened)</i> for the same firm in the one-year pre-ChatGPT period.
<i>GAI%-Initiative (Not Specific)</i>	Equals <i>GAI%-Initiative</i> if the GAI discussion does not contain specific GAI-related initiatives, and zero otherwise. We use ChatGPT-4 to identify these initiatives. Our prompt is as follows: Is the text that discusses real initiatives or plans in generative AI specific? Specificity includes mentioning a product, an application, a number, a function, or an example. Answer with 1 for “Yes”, 0 for “No”. Text: {}.
<i>ΔGAI%-Initiative (Not Specific)</i>	<i>GAI%-Initiative (Not Specific)</i> in post-ChatGPT period minus the average <i>GAI%-Initiative (Not Specific)</i> for the same firm in the one-year pre-ChatGPT period.
<i>GAI%-Initiative (No Partner)</i>	Equals <i>GAI%-Initiative</i> if the GAI discussion does not contain GAI-related initiatives that relate to a specific business partner, and zero otherwise. We use ChatGPT-4 to identify these initiatives. Our prompt is as follows: Is the text that discusses real initiatives or plans in generative AI specific? Specificity includes mentioning a product, an application, a number, a function, or an example. Answer with 1 for “Yes”, 0 for “No”. Text: {}.
<i>ΔGAI%-Initiative (No Partner)</i>	<i>GAI%-Initiative (No Partner)</i> in post-ChatGPT period minus the average <i>GAI%-Initiative (No Partner)</i> for the same firm in the one-year pre-ChatGPT period.
<i>GAI%-Initiative (Not Certain)</i>	Equals <i>GAI%-Initiative</i> if the GAI discussion of GAI-related initiatives is not sufficiently certain, and zero otherwise. We use ChatGPT-4 to identify these initiatives. Our prompt is as follows: Does the text that involves real initiatives or plans in generative AI discuss the initiatives or plans with sufficient certainty? Answer with 1 for “Yes”, 0 for “No”. Text: {}.
<i>ΔGAI%-Initiative (Not Certain)</i>	<i>GAI%-Initiative (Not Certain)</i> in post-ChatGPT period minus the average <i>GAI%-Initiative (Not Certain)</i> for the same firm in the one-year pre-ChatGPT period.
Market Reaction Variables	
<i> Call CAR </i>	Absolute cumulative abnormal returns over the two-day window [0, +1] around the conference call. Abnormal returns are calculated based on Fama-French five factor model. We multiply it by 100 for the ease of interpretation.
<i>Ab. Volume</i>	Excess trading volume over the two-day window [0, +1], measured as the two-day trading volume divided by number of shares outstanding, less the average two-day turnover in the estimation period [-120, -30], following Bushee et al. (2011). We multiply it by 100 for the ease of interpretation.
<i>GPT Launch CAR</i>	Abnormal return cumulated over a two-day window [0, +1], where day 0 is the ChatGPT launch date, November 30, 2022. We calculate abnormal return based on Fama-French five factor model, and multiply it by 100 for the ease of interpretation.

Appendix D (Continued)

Variable Definitions

Independent Variables	
<i>Abnormal Return Volatility</i>	Standard deviation of daily market adjusted return over the past year.
<i>Absolute Call Sentiment</i>	Percentage of sentiment words (Loughran and McDonald 2011) in the entire conference call transcript.
<i>Absolute Earnings Surprise</i>	Absolute value of the difference between quarterly EPS and consensus analyst forecasts (median) in the latest 90 days, scaled by the stock price at the end of the latest fiscal quarter.
<i>AIIE</i>	Industry-level labor exposure to AI from Felten et al. (2021), calculated as a weighted average of the AI Occupational Exposure (AIOE) scores across all occupations within an industry. AIOE for an occupation is derived by assessing the relevance of specific AI applications to the abilities required in that occupation.
<i>Analyst Coverage</i>	The natural logarithm of one plus the number of analysts who issue EPS forecasts for the firm over the past year.
<i>Customer</i>	Indicator variable that equals 1 if the firm is in the retail or service industries, and 0 otherwise. We classify retail industry by 2-digit SIC codes 52-59, and service industry by 2-digit SIC codes 72, 73, and 78.
<i>CS Threat</i>	Textual similarity between 10-K cybersecurity threats related sentences and the disclosures of actual incidents of cyberattacks from the Privacy Rights Clearinghouse's Data Breach Chronology Database. The firm's cybersecurity threats related sentences are extracted from "Item 1a. Risk Factors" from its latest 10-K filing, and identified as sentences containing keywords such as "cyberattack", "data breaches", and "data theft", following Florackis et al. (2023). We next compile the descriptions of cyberattack incidents on PRC database, and calculate its cosine similarity with the firm's cybersecurity threats related sentences.
<i>Firm Age</i>	Number of years the firm showing up in Compustat. We divide it by 10 for the ease of interpretation.
<i>High Tech Industry</i>	Indicator variable that equals 1 if 2-digit sic codes belong to 28, 35, 36, 73, and 87, and 0 otherwise.
<i>Institutional Shareholding</i>	Percentage of shares outstanding owned by institutional investors, at the end of the latest fiscal quarter.
<i>Market to Book</i>	Market value of equity, at the latest fiscal year end divided by book value of equity.
<i>Miss R&D</i>	Indicator variable that equals 1 when Compustat variable (xrd) is missing, and 0 otherwise.
<i>Patents</i>	The total number of patents filed over the past three calendar years. To overcome the skewness, we take inverse hyperbolic sine transformation following Glaeser (2018).
<i>Quarterly Market to Book</i>	Market value at the end of the latest fiscal quarter divided by book value.
<i>Quarterly Market Value</i>	Logarithm of market value at the end of the latest fiscal quarter.
<i>Quarterly Return on Asset</i>	Quarterly net income over total assets for the latest fiscal quarter.
<i>Quarterly Return Volatility</i>	Return volatility over the period of [-127, -2] days relative to the conference call date (i.e., day 0) (Mayew and Venkatachalam 2012).
<i>R&D</i>	R&D expenditure over total assets for the latest fiscal quarter.
<i>Size</i>	The natural logarithm of total assets, at the latest fiscal quarter end.
<i>Trade Secret</i>	An indicator equals 1 if the firm's past three year 10-K filings mention "trade secret" or "trade secrecy," and 0 otherwise.

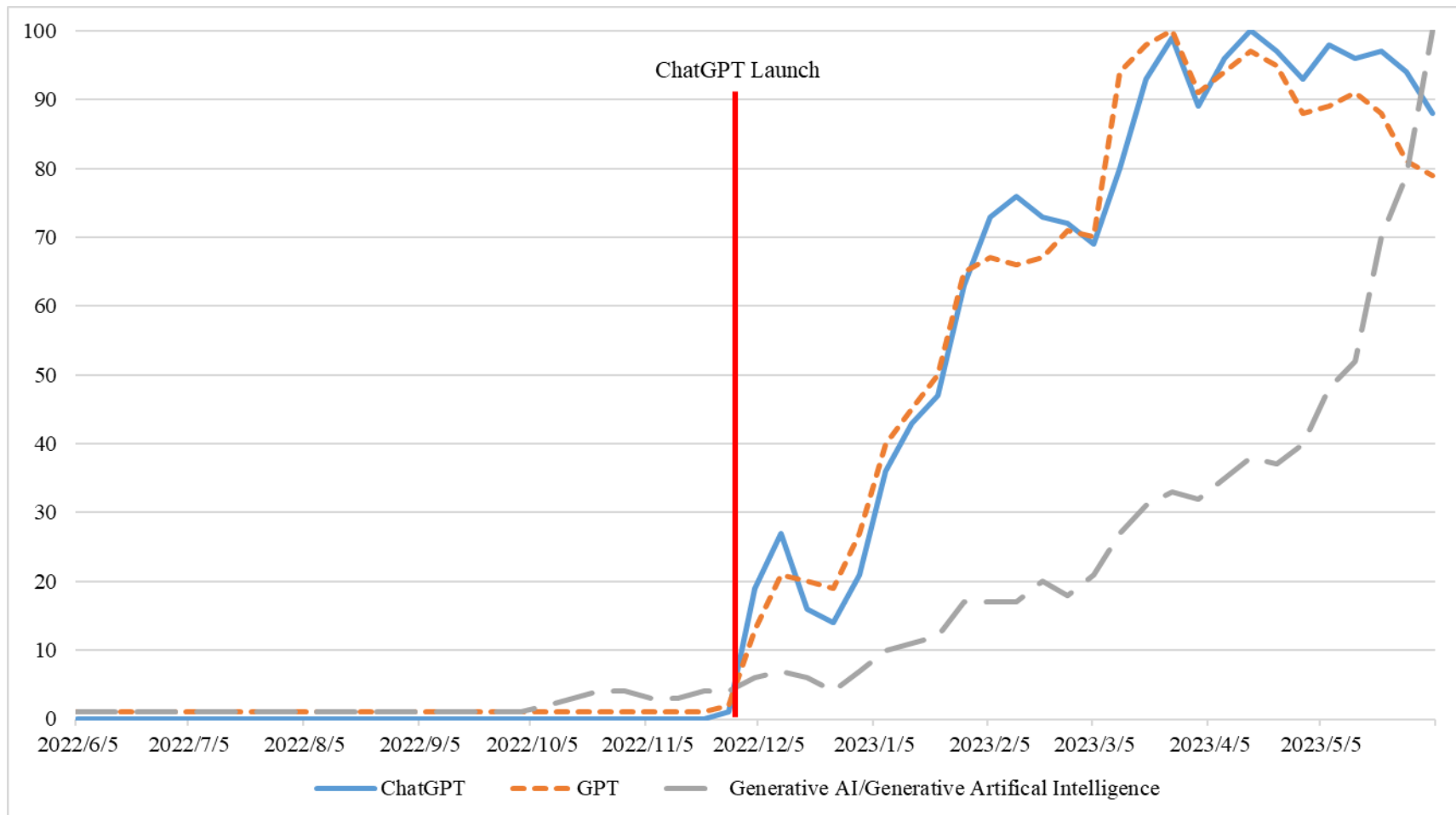


Figure 1 Google Search of GAI-related Keywords Over Time.

This figure presents the time-series trend of worldwide Google search interest index for the phrase “ChatGPT”, “GPT”, and “Generative AI/Generative Artificial Intelligence” between June 2022 and June 2023.

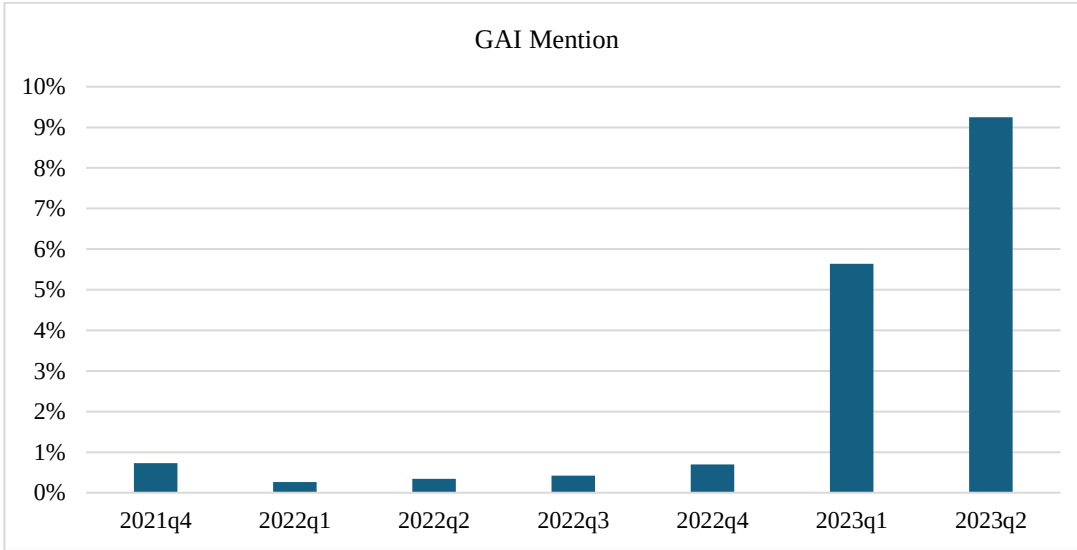


Figure 2A: Percentetage of calls mentioning GAI-related keywords by quarter

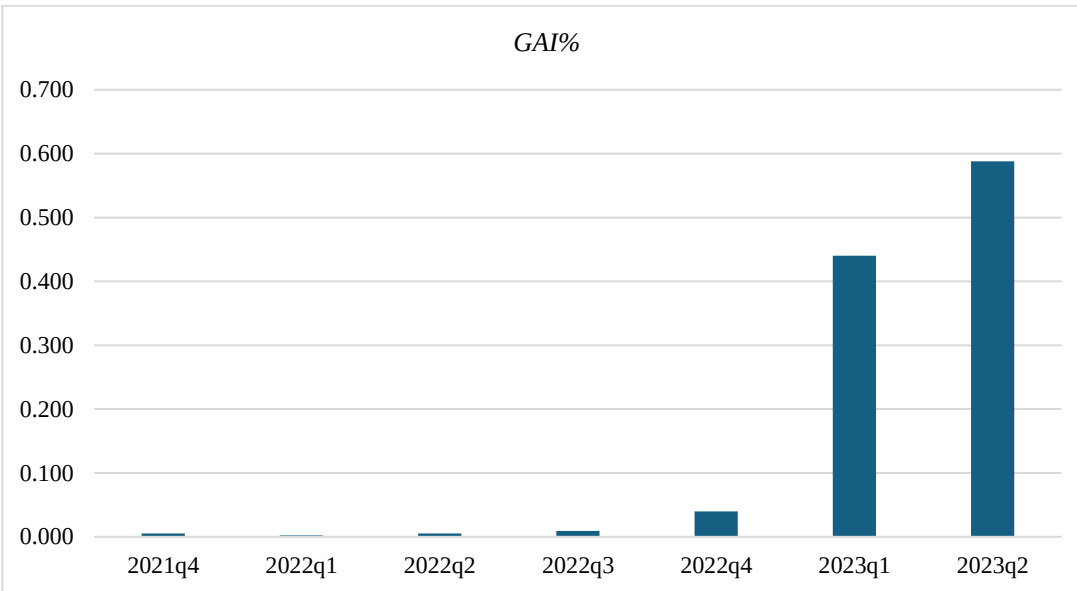


Figure 2B: Mean of *GAI%* by quarter

Figure 2 GAI Discussions in Conference Calls Over Time

Figure 2A presents the percentage of conference calls in which managers discuss GAI for each quarter between 2021Q4 and 2023Q2. Figure 2B presents the mean of the measure of managers' GAI discussion intensity *GAI%* (defined in Appendix D) for each quarter between 2021Q4 and 2023Q2.

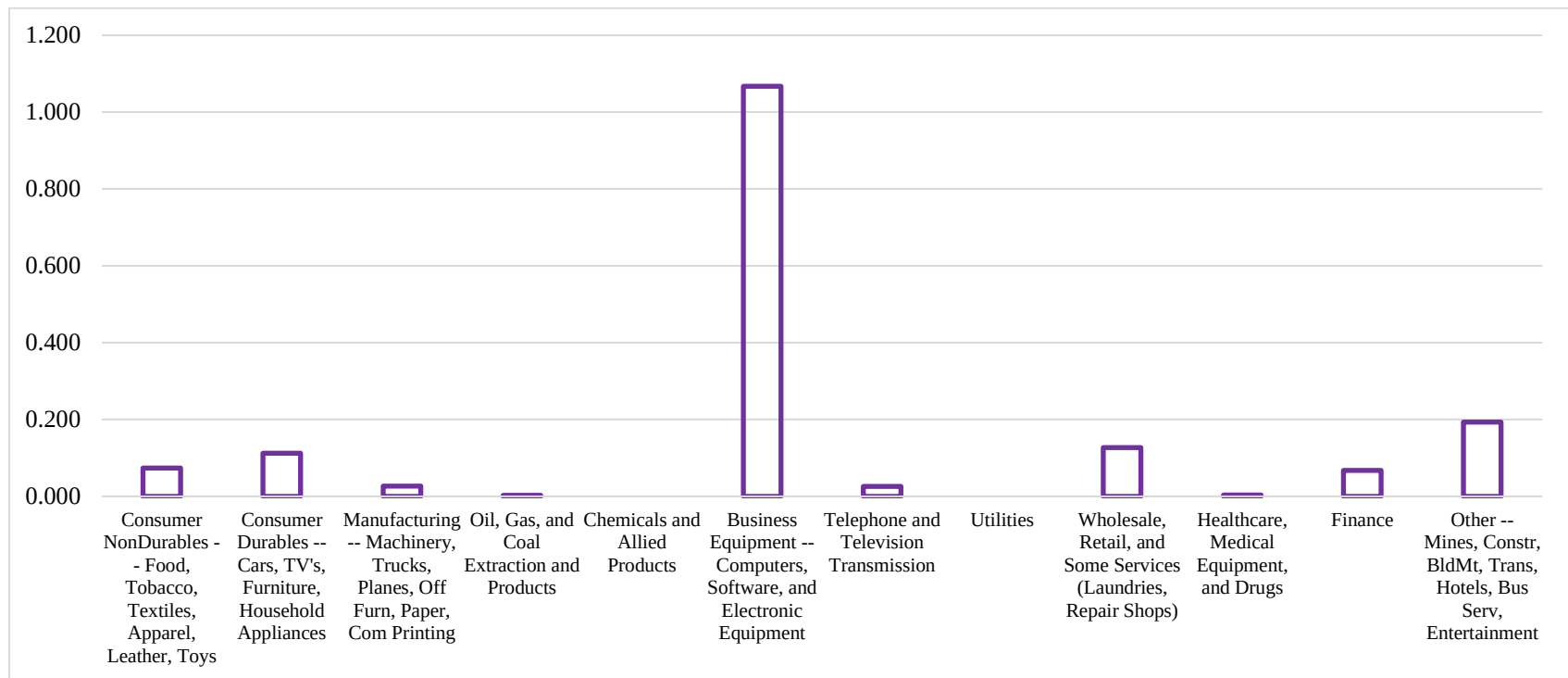


Figure 3A: Change of managers' GAI discussion intensity

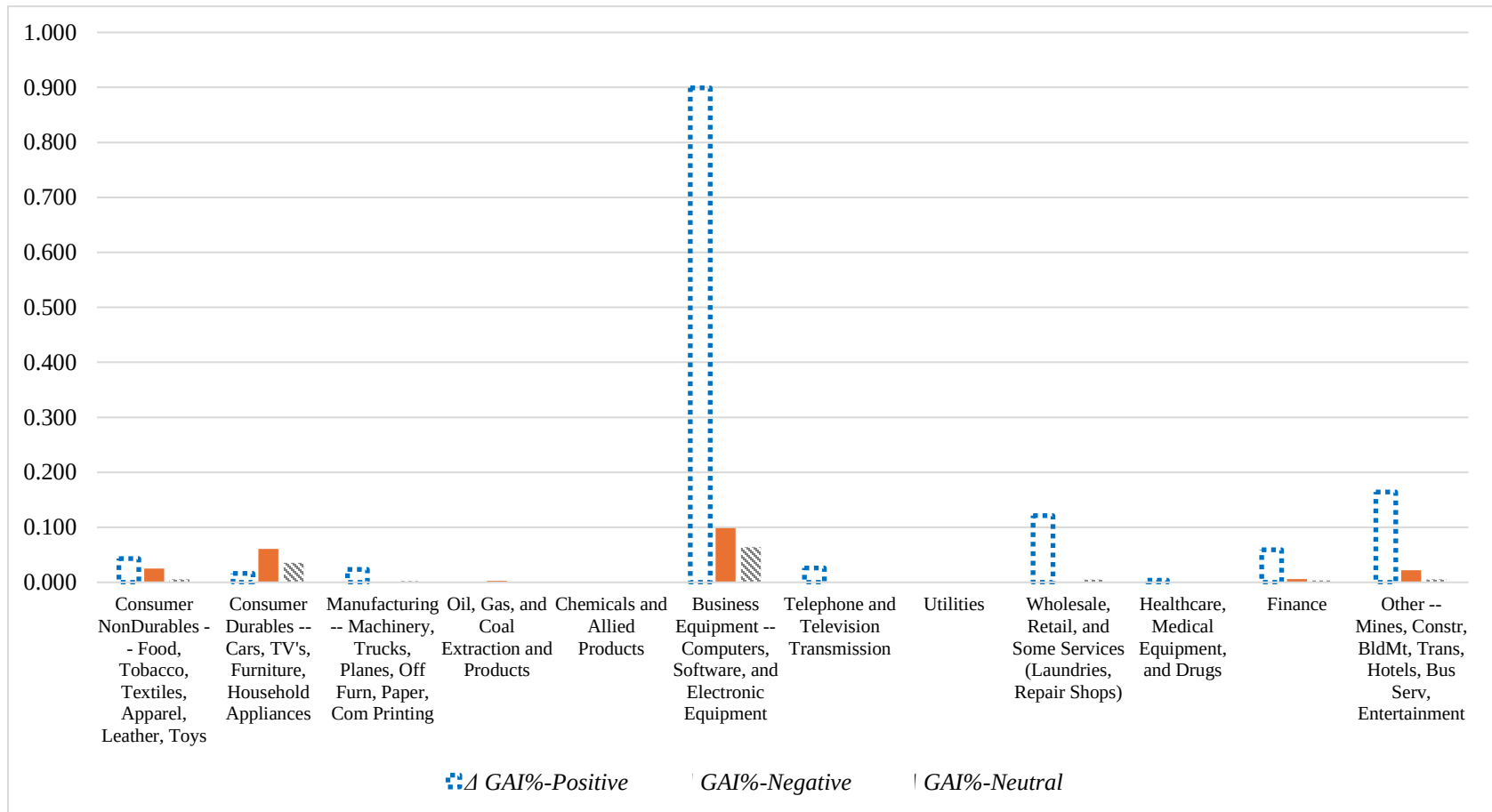


Figure 3B: Changes of managers' GAI discussions with positive, negative, and neutral tones

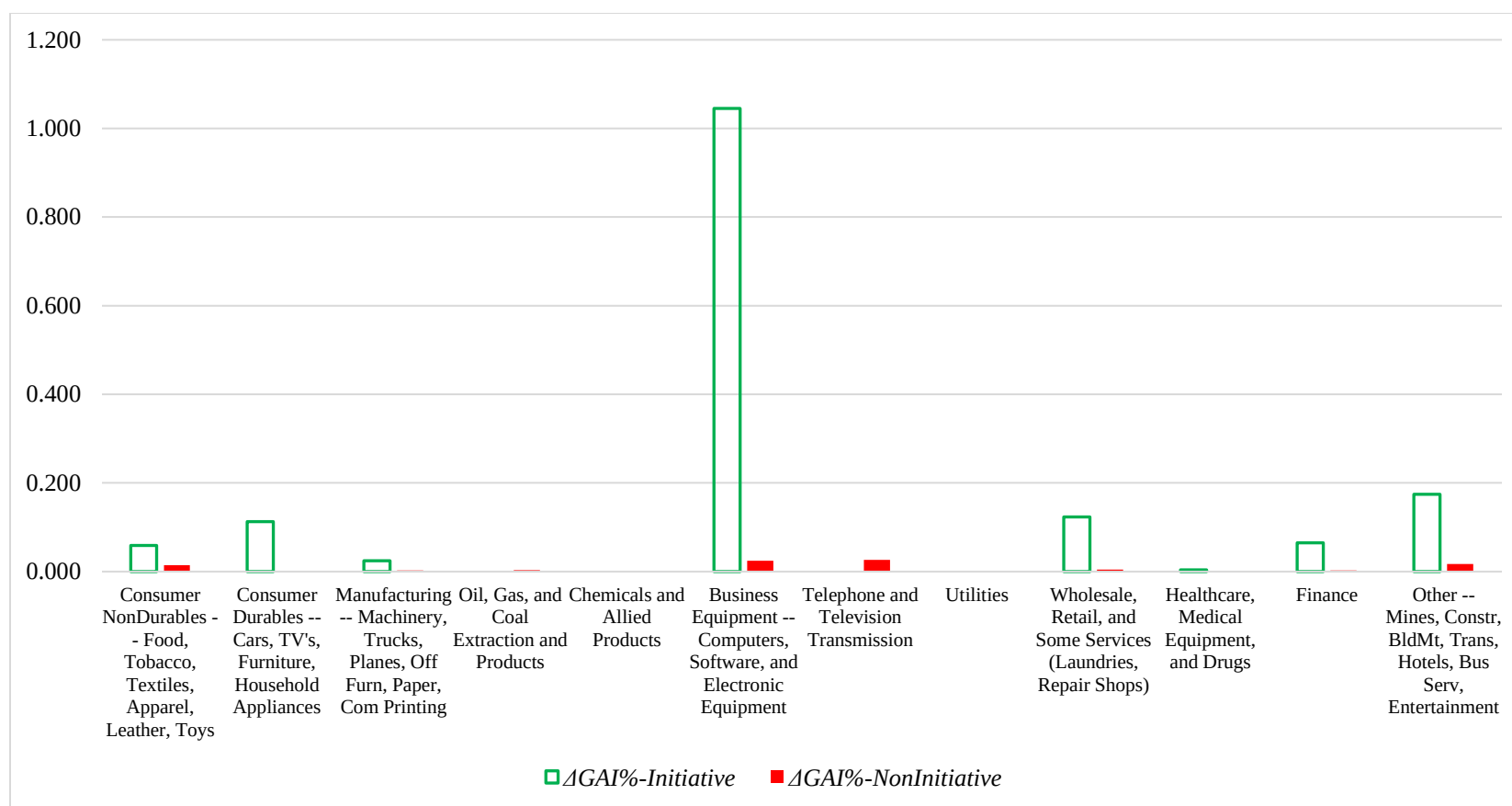


Figure 3C: Changes of managers' GAI discussions with and without initiatives

Figure 3 Changes of Manager GAI Discussions across Industries

Figure 3A presents average change of managers' GAI discussion intensity in conference calls ($\Delta GAI\%$) in each of the Fama-French 12 industries. Figure 3B separately presents the results for positive-, negative-, and neutral-tone discussions ($\Delta GAI\% - \text{Pos}$, $\Delta GAI\% - \text{Neg}$, and $\Delta GAI\% - \text{Neu}$). Figure 3C separately presents the results for discussions with and without action initiatives ($\Delta GAI\% - \text{Initiative}$ and $\Delta GAI\% - \text{NonInitiative}$). Variable definitions are in Appendix D.

Table 1 Summary Statistics**Panel A: GAI Discussion Variables**

	<i>N</i>	Mean	SD	P25	P50	P75
Manager Discussions						
<i>GAI%</i>	6,192	0.316	1.667	0.000	0.000	0.000
$\Delta GAI\%$	6,192	0.306	1.609	0.000	0.000	0.000
<i>GAI%-Pos</i>	6,192	0.263	1.517	0.000	0.000	0.000
$\Delta GAI\%-Pos$	6,192	0.257	1.477	0.000	0.000	0.000
<i>GAI%-Neg</i>	6,192	0.030	0.542	0.000	0.000	0.000
$\Delta GAI\%-Neg$	6,192	0.030	0.542	0.000	0.000	0.000
<i>GAI%-Neu</i>	6,192	0.019	0.341	0.000	0.000	0.000
$\Delta GAI\%-Neu$	6,192	0.018	0.341	0.000	0.000	0.000
<i>GAI%-Initiative</i>	6,192	0.305	1.655	0.000	0.000	0.000
$\Delta GAI\%-Initiative$	6,192	0.297	1.604	0.000	0.000	0.000
<i>GAI%-NonInitiative</i>	6,192	0.012	0.219	0.000	0.000	0.000
$\Delta GAI\%-NonInitiative$	6,192	0.010	0.220	0.000	0.000	0.000
$\Delta GAI\%-Positive-Broad$	6,192	0.255	1.478	0.000	0.000	0.000
$\Delta GAI\%-Initiative$ (Already Happened)	6,192	0.279	1.578	0.000	0.000	0.000
$\Delta GAI\%-Initiative$ (Specific)	6,192	0.251	1.511	0.000	0.000	0.000
$\Delta GAI\%-Initiative$ (Partner)	6,192	0.116	0.881	0.000	0.000	0.000
$\Delta GAI\%-Initiative$ (Certain)	6,192	0.209	1.284	0.000	0.000	0.000
$\Delta GAI\%-Initiative$ (Not Already Happened)	6,192	0.018	0.306	0.000	0.000	0.000
$\Delta GAI\%-Initiative$ (Not Specific)	6,192	0.012	0.107	0.000	0.000	0.000
$\Delta GAI\%-Initiative$ (No Partner)	6,192	0.107	0.696	0.000	0.000	0.000
$\Delta GAI\%-Initiative$ (Not Certain)	6,192	0.029	0.236	0.000	0.000	0.000
Analyst Discussions						
<i>GAI%-Analyst</i>	6,192	0.318	2.113	0.000	0.000	0.000
$\Delta GAI\%-Analyst$	6,192	0.318	2.112	0.000	0.000	0.000
<i>GAI%-Pos-Analyst</i>	6,192	0.168	1.346	0.000	0.000	0.000
$\Delta GAI\%-Pos-Analyst$	6,192	0.168	1.346	0.000	0.000	0.000
<i>GAI%-Neg-Analyst</i>	6,192	0.073	1.790	0.000	0.000	0.000
$\Delta GAI\%-Neg-Analyst$	6,192	0.069	1.799	0.000	0.000	0.000
<i>GAI%-Neu-Analyst</i>	6,192	0.049	0.750	0.000	0.000	0.000
$\Delta GAI\%-Neu-Analyst$	6,192	0.048	0.750	0.000	0.000	0.000
<i>GAI%-Initiative-Analyst</i>	6,192	0.212	1.620	0.000	0.000	0.000
$\Delta GAI\%-Initiative-Analyst$	6,192	0.211	1.613	0.000	0.000	0.000
<i>GAI%-NonInitiative-Analyst</i>	6,192	0.090	1.452	0.000	0.000	0.000
$\Delta GAI\%-NonInitiative-Analyst$	6,192	0.089	1.446	0.000	0.000	0.000

Table 1 (Continued)**Panel B: Other Variables**

	<i>N</i>	Mean	SD	P25	P50	P75
Independent Variables of Interest						
<i>Patents</i> (untransformed)	6,192	32.201	153.295	0.000	0.000	2.000
<i>Patents</i> (inverse hyperbolic sine)	6,192	1.067	1.904	0.000	0.000	1.444
<i>Trade Secret</i>	6,192	0.829	0.377	1.000	1.000	1.000
<i>R&D</i>	6,192	0.064	0.113	0.000	0.012	0.081
<i>Miss R&D</i>	6,192	0.331	0.471	0.000	0.000	1.000
<i>CS Threat</i>	6,192	0.691	0.083	0.642	0.695	0.755
<i>AIIE</i>	6,192	0.642	0.962	-0.075	0.464	1.705
<i>Customer</i>	6,192	0.204	0.403	0.000	0.000	0.000
Control Variables						
<i>Size</i> (unlogged, \$million)	6,192	15,852	33,091	910	3,165	12,470
<i>Size</i>	6,192	8.081	1.942	6.814	8.060	9.431
<i>Market to Book</i>	6,192	6.449	14.473	1.501	3.030	6.452
<i>Firm Age</i> (in years)	6,192	27.123	20.071	10.000	24.000	36.000
<i>High Tech Industry</i>	6,192	0.486	0.500	0.000	0.000	1.000
<i>Analyst Coverage</i> (unlogged)	6,192	13.122	9.691	5.000	11.000	19.000
<i>Analyst Coverage</i>	6,192	2.362	0.838	1.792	2.485	2.996
<i>Institutional Shareholding</i>	6,192	0.760	0.210	0.670	0.822	0.912
<i>Abnormal Return Volatility</i>	6,192	0.026	0.016	0.016	0.022	0.032
<i>Quarterly Market Value</i>	6,187	8.058	2.151	6.601	8.096	9.563
<i>Quarterly Market to Book</i>	6,187	3.619	15.518	1.128	2.297	4.825
<i>Quarterly Return Volatility</i>	6,192	0.030	0.015	0.019	0.025	0.036
<i>Absolute Earnings Surprise</i>	6,191	0.030	0.081	0.002	0.006	0.019
<i>Absolute Call Sentiment</i>	6,185	2.667	0.578	2.274	2.646	3.024
<i>Quarterly Return on Asset</i>	6,192	-0.009	0.060	-0.012	0.006	0.018
Market Reaction Variables						
<i> Call CAR </i>	6,181	5.739	6.672	1.355	3.419	7.581
<i>Ab. Volume</i>	6,181	1.535	3.379	-0.026	0.578	1.768
<i>GPT Launch CAR</i>	2,869	-0.595	4.509	-2.529	-0.221	1.595

This table reports summary statistics of the variables used in the analysis. Panel A presents variables of GAI discussions in conference calls; Panel B presents other variables. Variable definitions are in Appendix D.

Table 2 Univariate Comparison of Changes in Managers' GAI Discussions**Panel A: Partition by High-tech Industries**

	$\Delta GAI\%$	$\Delta GAI\%-Pos$	$\Delta GAI\%-Neg$	$\Delta GAI\%-Neu$	$\Delta GAI\%-Initiative$	$\Delta GAI\%-NonInitiative$
High Tech Industries	0.565	0.471	0.055	0.036	0.553	0.014
Other Industries	0.061	0.054	0.006	0.001	0.055	0.006
Diff	0.504***	0.417***	0.049***	0.035***	0.498***	0.008

Panel B: Partition by Firm Size

	$\Delta GAI\%$	$\Delta GAI\%-Pos$	$\Delta GAI\%-Neg$	$\Delta GAI\%-Neu$	$\Delta GAI\%-Initiative$	$\Delta GAI\%-NonInitiative$
Small Firms	0.257	0.222	0.022	0.009	0.248	0.009
Large Firms	0.355	0.291	0.038	0.027	0.346	0.010
Diff	-0.098**	-0.069*	-0.016	-0.018**	-0.098**	-0.001

Panel C: Partition by Firm Age

	$\Delta GAI\%$	$\Delta GAI\%-Pos$	$\Delta GAI\%-Neg$	$\Delta GAI\%-Neu$	$\Delta GAI\%-Initiative$	$\Delta GAI\%-NonInitiative$
Young Firms	0.376	0.305	0.042	0.028	0.363	0.014
Mature Firms	0.227	0.203	0.015	0.007	0.223	0.004
Diff	0.149***	0.102***	0.027*	0.021**	0.140***	0.010*

Panel D: Partition by Market to Book Ratio

	$\Delta GAI\%$	$\Delta GAI\%-Pos$	$\Delta GAI\%-Neg$	$\Delta GAI\%-Neu$	$\Delta GAI\%-Initiative$	$\Delta GAI\%-NonInitiative$
Growth Firms	0.512	0.433	0.046	0.029	0.497	0.015
Value Firms	0.100	0.080	0.013	0.007	0.097	0.004
Diff	0.412***	0.353***	0.033**	0.022**	0.400***	0.011**

This table reports the univariate comparison of changes of managers' GAI discussions, overall intensity and those with different features (sentiment and initiative-relatedness) based on industry nature (high-tech industries or not) and fundamental firm characteristics (size, age, and market-to-book) partitioned based on the sample median. Variable definitions are in Appendix D. ***, **, and * indicate two-sided significance at the 1%, 5%, and 10% levels, respectively.

Table 3 Determinants of Changes in GAI Discussions in Conference Calls**Panel A: Manager Discussions**

Dependent Variable	$\Delta GAI\%$							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Patents</i>		0.04* (1.95)						0.06** (2.30)
<i>Trade Secret</i>			0.04* (1.93)					0.00 (0.09)
<i>R&D</i>				-0.02 (-0.38)				0.05 (0.99)
<i>Miss R&D</i>				-0.04 (-1.02)				0.00 (0.09)
<i>CS Threat</i>					0.04 (1.56)			0.05** (2.25)
<i>AIIE</i>						0.10* (1.76)		0.07*** (2.87)
<i>Customer</i>							0.48*** (3.17)	0.51*** (3.94)
<i>Size</i>	0.04*** (2.74)	0.04*** (2.77)	0.04*** (2.77)	0.04** (2.26)	0.05*** (2.87)	0.03 (1.25)	0.04** (2.46)	0.04* (1.82)
<i>Market to Book</i>	0.01*** (2.64)	0.01*** (2.67)	0.01*** (2.62)	0.01*** (2.69)	0.01** (2.55)	0.01** (2.51)	0.01** (2.58)	0.00** (2.44)
<i>Firm Age</i>	-0.04** (-2.24)	-0.05** (-2.39)	-0.04** (-2.21)	-0.04** (-2.22)	-0.04** (-2.34)	-0.02** (-2.17)	-0.03*** (-3.20)	-0.02** (-2.06)
<i>High Tech Industry</i>	0.29* (1.93)	0.27* (1.80)	0.28* (1.89)	0.29** (2.17)	0.27* (1.81)	0.23* (1.97)	0.20** (2.51)	0.08 (1.53)
<i>Analyst Coverage</i>	0.08 (1.49)	0.06 (1.29)	0.08 (1.44)	0.08* (1.70)	0.07 (1.44)	0.09 (1.59)	0.07 (1.47)	0.03 (1.22)
<i>Institutional Shareholding</i>	-0.35* (-1.85)	-0.33* (-1.83)	-0.36* (-1.89)	-0.35* (-1.80)	-0.37* (-1.94)	-0.30 (-1.62)	-0.39* (-1.93)	-0.37* (-1.86)
<i>Abnormal Return Volatility</i>	-1.88 (-1.02)	-1.78 (-0.95)	-2.00 (-1.10)	-1.67 (-1.06)	-1.75 (-0.97)	-1.21 (-0.82)	-1.48 (-1.29)	-1.40 (-1.04)
Observations	6,192	6,192	6,192	6,192	6,192	6,192	6,192	6,192
R-squared	0.05	0.05	0.05	0.05	0.05	0.06	0.09	0.10

Table 3 (Continued)**Panel B: Analyst Discussions**

Dependent Variable	<i>ΔGAI%-Analyst</i>							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Patents</i>		-0.02 (-0.96)						-0.01 (-0.33)
<i>Trade Secret</i>			-0.01 (-0.57)					-0.03 (-0.91)
<i>R&D</i>				-0.02 (-0.54)				0.05 (0.96)
<i>Miss R&D</i>				-0.03 (-0.68)				-0.03 (-0.61)
<i>CS Threat</i>					0.02 (0.83)			0.04*** (2.86)
<i>AIIE</i>						0.10* (1.77)		0.06*** (3.07)
<i>Customer</i>							0.48*** (4.07)	0.49*** (4.75)
<i>Size</i>	0.02 (1.17)	0.02 (1.30)	0.02 (1.15)	0.01 (1.30)	0.02 (1.27)	-0.00 (-0.01)	0.02 (1.00)	0.02* (1.68)
<i>Market to Book</i>	0.00*** (2.65)	0.00*** (2.62)	0.00*** (2.66)	0.00*** (2.73)	0.00** (2.56)	0.00** (2.50)	0.00** (2.58)	0.00** (2.30)
<i>Firm Age</i>	-0.04* (-1.94)	-0.03** (-2.04)	-0.04* (-1.94)	-0.04* (-1.93)	-0.04** (-2.00)	-0.02* (-1.95)	-0.02*** (-2.83)	-0.01** (-2.00)
<i>High Tech Industry</i>	0.24* (1.92)	0.25* (1.90)	0.25* (1.92)	0.25** (2.24)	0.24* (1.82)	0.19** (2.03)	0.15*** (2.94)	0.07* (1.72)
<i>Analyst Coverage</i>	0.07 (1.35)	0.08 (1.36)	0.07 (1.36)	0.07 (1.58)	0.07 (1.32)	0.08 (1.44)	0.06 (1.33)	0.04 (1.46)
<i>Institutional Shareholding</i>	-0.21* (-1.95)	-0.22* (-1.93)	-0.21* (-1.94)	-0.21* (-1.85)	-0.22** (-2.04)	-0.17* (-1.66)	-0.25** (-2.07)	-0.25* (-1.95)
<i>Abnormal Return Volatility</i>	-2.49 (-1.32)	-2.53 (-1.33)	-2.44 (-1.32)	-2.21 (-1.23)	-2.43 (-1.31)	-1.87 (-1.23)	-2.08 (-1.61)	-2.16 (-1.33)
Observations	6,192	6,192	6,192	6,192	6,192	6,192	6,192	6,192
R-squared	0.03	0.03	0.03	0.03	0.03	0.04	0.07	0.07

This table reports the regression results of the determinants of changes in managers' GAI discussions after the launch of ChatGPT (Panel A), as well as those of analysts (Panel B). Variable definitions are in Appendix D. The dependent variable and key non-indicator independent variables *Patents*, *R&D*, *CS Threat*, and *AIIE* are standardized. *t*-statistics (in parentheses) are based on standard errors clustered at the industry level. ***, **, and * indicate two-sided significance at the 1%, 5%, and 10% levels, respectively.

Table 4 Determinants of Changes in GAI Discussions with Different Tones**Panel A: Manager Discussions**

Dependent Variable	$\Delta GAI\%-Pos$	$\Delta GAI\%-Neg$	$\Delta GAI\%-Neu$
	(1)	(2)	(3)
<i>Patents</i>	0.06** (2.14)	-0.01 (-0.24)	0.01 (0.82)
<i>Trade Secret</i>	0.00 (0.19)	-0.00 (-0.03)	-0.01 (-0.40)
<i>R&D</i>	0.05 (1.04)	-0.00 (-0.56)	-0.00 (-0.06)
<i>Miss R&D</i>	0.01 (0.29)	-0.01 (-0.68)	-0.02 (-0.74)
<i>CS Threat</i>	0.05** (2.48)	0.01 (0.87)	0.01 (0.68)
<i>AIIE</i>	0.06*** (2.71)	0.02** (2.48)	0.02** (2.43)
<i>Customer</i>	0.48*** (4.19)	0.12** (2.49)	0.13** (2.57)
Control	Yes	Yes	Yes
Observations	6,192	6,192	6,192
R-squared	0.08	0.01	0.01
Diff in Coefficients in Columns (1) and (2)			<i>p</i> -Value for Diff >0
<i>Patents</i>	0.07		0.07*
<i>Trade Secret</i>	-0.00		0.41
<i>R&D</i>	0.05		0.13
<i>CS Threat</i>	0.04		<0.01***
<i>AIIE</i>	0.04		<0.01***
<i>Customer</i>	0.36		<0.01***

Table 4 (Continued)

Panel B: Analyst Discussions			
Dependent Variable	$\Delta GAI\%-Pos-Analyst$	$\Delta GAI\%-Neg-Analyst$	$\Delta GAI\%-Neu-Analyst$
	(1)	(2)	(3)
<i>Patents</i>	-0.01 (-0.27)	-0.00 (-0.06)	0.00 (0.31)
<i>Trade Secret</i>	-0.02 (-0.85)	-0.01 (-0.93)	-0.03 (-0.73)
<i>R&D</i>	0.06 (1.01)	-0.01 (-1.00)	-0.02 (-1.03)
<i>Miss R&D</i>	-0.01 (-0.11)	-0.02* (-1.68)	-0.03 (-1.30)
<i>CS Threat</i>	0.05*** (2.98)	-0.00 (-0.26)	-0.00 (-0.27)
<i>AIIE</i>	0.05*** (3.27)	0.02* (1.92)	0.02** (2.56)
<i>Customer</i>	0.41*** (4.80)	0.12*** (3.18)	0.19*** (5.11)
Control	Yes	Yes	Yes
Observations	6,192	6,192	6,192
R-squared	0.05	0.00	0.01
		Diff in Coefficients in Columns (1) and (2)	p-Value for Diff >0
<i>Patents</i>		-0.01	0.44
<i>Trade Secret</i>		-0.01	0.29
<i>R&D</i>		0.07	0.15
<i>CS Threat</i>		0.05	0.02**
<i>AIIE</i>		0.03	<0.01***
<i>Customer</i>		0.29	<0.01***

This table reports the regression results of the determinants of changes in managers' discussions of GAI after the launch of ChatGPT (in Panel A) as well as those of analysts' (in Panel B) based on the discussion sentiment (positive, negative, or neutral). Variable definitions are in Appendix D. The dependent variable and key non-indicator independent variables *Patents*, *R&D*, *CS Threat*, and *AIIE* are standardized. *t*-statistics (in parentheses) are based on standard errors clustered at the industry level. ***, **, and * indicate two-sided significance at the 1%, 5%, and 10% levels, respectively.

Table 5 Determinants of Changes in GAI Discussions with and without Initiatives**Panel A: Manager Discussions**

Dependent Variable	$\Delta GAI\%-Initiative$ (1)	$\Delta GAI\%-NonInitiative$ (2)
<i>Patents</i>	0.06** (2.30)	0.00 (0.13)
<i>Trade Secret</i>	-0.00 (-0.00)	0.02 (0.88)
<i>R&D</i>	0.05 (1.04)	-0.04 (-1.34)
<i>Miss R&D</i>	0.01 (0.30)	-0.08 (-1.56)
<i>CS Threat</i>	0.06** (2.52)	-0.03 (-1.21)
<i>AIIE</i>	0.07*** (2.75)	0.03*** (3.47)
<i>Customer</i>	0.51*** (3.98)	0.05 (1.63)
Control	Yes	Yes
Observations	6,192	6,192
R-squared	0.10	0.00
Diff in Coefficients in Columns (1) and (2)		<i>p</i> -Value for Diff >0
<i>Patents</i>	0.06	0.06*
<i>Trade Secret</i>	-0.02	0.28
<i>R&D</i>	0.09	0.11
<i>CS Threat</i>	0.09	<0.01***
<i>AIIE</i>	0.04	0.08*
<i>Customer</i>	0.46	<0.01***

Table 5 (Continued)**Panel B: Analyst Discussions**

Dependent Variable	$\Delta GAI\%-Initiative-Analyst$	$\Delta GAI\%-NonInitiative-Analyst$
	(1)	(2)
<i>Patents</i>	-0.02 (-0.82)	0.01 (0.93)
<i>Trade Secret</i>	-0.03 (-0.79)	-0.03 (-1.16)
<i>R&D</i>	0.05 (1.03)	0.02 (0.76)
<i>Miss R&D</i>	-0.03 (-0.48)	-0.01 (-0.28)
<i>CS Threat</i>	0.05*** (2.62)	0.01 (1.29)
<i>AIIE</i>	0.06*** (3.18)	0.02** (2.35)
<i>Customer</i>	0.44*** (4.98)	0.20*** (3.39)
Control	Yes	Yes
Observations	6,192	6,192
R-squared	0.06	0.01
	Diff in Coefficients in Columns (1) and (2)	<i>p</i> -Value for Diff >0
<i>Patents</i>	-0.03	0.16
<i>Trade Secret</i>	-0.00	0.43
<i>R&D</i>	0.03	0.11
<i>CS Threat</i>	0.04	0.01**
<i>AIIE</i>	0.04	<0.01***
<i>Customer</i>	0.24	<0.01***

This table reports the regression results of changes in managers' discussion of GAI after the launch of ChatGPT (in Panel A) as well as those of analysts' (in Panel B) on innovation intensity, cybersecurity threats, product differentiation, labor, and customer operations, respectively, based on the discussion content (initiative vs. non-initiative). Please see Appendix D for variable definitions. The dependent variables $\Delta GAI\%-Initiative$ and $\Delta GAI\%-NonInitiative$ and key non-indicator independent variables *Patents*, *R&D*, *CS Threat*, and *AIIE* are standardized. Panel A focuses on managers' discussions, whereas Panel B focuses on analysts' discussions. *t*-statistics (in parentheses) are based on standard errors clustered at the industry level (SIC3). ***, **, and * indicate two-sided significance at the 1%, 5%, and 10% levels, respectively.

Table 6 Analysis of AI-Related Job Postings

	GAI Discussion Firms (1)	Initiative GAI Discussion Firms (2)	No GAI Discussion Firms (3)	Diff (1) - (3)	Diff (2) - (3)
# of AI-related Postings	365	376	103	262***	273***
% of AI-related Postings	26.59%	26.69%	11.18%	15.41%***	15.51%***
<i>N</i>	200	174	1,825		

This table reports firms' AI-related job postings in one year after the launch of ChatGPT (from Dec 2022 Dec to Nov 2023) for firms with GAI discussions, firms with initiative-related GAI discussions, and firms without discussions. ***, **, and * indicate two-sided significance at the 1%, 5%, and 10% levels, respectively.

Table 7 Market Reaction to Conference Calls**Panel A: Full Sample**

Dependent Variable	Call CAR (1)	Ab. Volume (2)	Call CAR (3)	Ab. Volume (4)	Call CAR (5)	Ab. Volume (6)
$\Delta GAI\%$	0.03*** (5.67)	0.05*** (3.90)				
$\Delta GAI\%$ -Pos-Broad			0.03*** (5.11)	0.05*** (3.78)		
$\Delta GAI\%$ -Initiative					0.03*** (6.15)	0.05*** (3.93)
$\Delta GAI\%$ -NonInitiative					-0.00 (-0.07)	0.00 (0.36)
Quarterly MV	-0.10*** (-15.25)	-0.00 (-0.02)	-0.10*** (-15.24)	-0.00 (-0.00)	-0.10*** (-15.24)	-0.00 (-0.02)
Quarterly Market to Book	0.00 (1.39)	0.00* (1.78)	0.00 (1.41)	0.00* (1.79)	0.00 (1.40)	0.00* (1.78)
Quarterly Return Volatility	15.97*** (9.59)	17.60*** (7.38)	16.01*** (9.59)	17.60*** (7.39)	15.98*** (9.59)	17.60*** (7.37)
Absolute Earnings Surprise	-0.23 (-1.19)	-0.73* (-1.83)	-0.23 (-1.18)	-0.73* (-1.83)	-0.23 (-1.19)	-0.73* (-1.83)
Absolute Call Sentiment	0.24*** (9.28)	0.21*** (6.01)	0.24*** (9.31)	0.21*** (6.05)	0.24*** (9.29)	0.21*** (6.01)
Quarterly Return on Asset	0.14 (0.60)	0.60* (1.72)	0.13 (0.60)	0.60* (1.72)	0.14 (0.61)	0.60* (1.73)
Observations	6,168	6,168	6,168	6,168	6,168	6,168
R-squared	0.18	0.07	0.18	0.07	0.18	0.07

Panel B: Low Institutional Shareholding

Dependent Variable	Call CAR (1)	Ab. Volume (2)	Call CAR (3)	Ab. Volume (4)	Call CAR (5)	Ab. Volume (6)
$\Delta GAI\%$	0.01* (1.90)	0.03* (1.74)				
$\Delta GAI\%$ -Pos-Broad			0.01 (1.09)	0.03 (1.62)		
$\Delta GAI\%$ -Initiative					0.01** (2.01)	0.03* (1.75)
$\Delta GAI\%$ -NonInitiative					-0.02** (-2.09)	0.02 (1.14)
Control	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,055	2,055	2,055	2,055	2,055	2,055
R-squared	0.15	0.06	0.15	0.06	0.15	0.06

Table 7 (Continued)**Panel C: High Institutional Shareholding**

Dependent Variable	<i>Call CAR</i> (1)	<i>Ab. Volume</i> (2)	<i>Call CAR</i> (3)	<i>Ab. Volume</i> (4)	<i>Call CAR</i> (5)	<i>Ab. Volume</i> (6)
$\Delta GAI\%$	0.05*** (5.83) [<0.01]	0.08*** (5.72) [0.02]				
$\Delta GAI\% - Pos - Broad$			0.05*** (4.32) [0.05]	0.09*** (5.68) [0.02]		
$\Delta GAI\% - Initiative$					0.05*** (5.51) [<0.01]	0.09*** (5.33) [0.03]
$\Delta GAI\% - NonInitiative$					0.01 (0.26) [0.22]	0.00 (0.22) [0.46]
Control	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,056	2,056	2,056	2,056	2,056	2,056
R-squared	0.16	0.09	0.16	0.09	0.16	0.09

Table 7 (Continued)**Panel D: Initiative Types**

Dependent Variable	Call CAR (1)	Ab. Volume (2)	Call CAR (3)	Ab. Volume (4)	Call CAR (5)	Ab. Volume (6)	Call CAR (7)	Ab. Volume (8)
<i>ΔGAI%-Initiative (Already Happened)</i>	0.03*** (5.62)	0.05*** (4.00)						
<i>ΔGAI%-Initiative (Specific)</i>			0.04*** (5.60)	0.05*** (3.76)				
<i>ΔGAI%-Initiative (Partner)</i>					0.03*** (7.42)	0.05*** (4.82)		
<i>ΔGAI%-Initiative (Certain)</i>							0.04*** (9.07)	0.06*** (4.25)
<i>ΔGAI%-Initiative (Not Already Happened)</i>	0.02** (1.99)	0.01 (1.33)						
<i>ΔGAI%-Initiative (Not Specific)</i>			0.01 (1.03)	0.02 (1.28)				
<i>ΔGAI%-Initiative (No Partner)</i>					0.02** (2.47)	0.03** (2.47)		
<i>ΔGAI%-Initiative (Not Certain)</i>							0.00 (0.54)	0.01 (0.65)
<i>ΔGAI%-NonInitiative</i>	-0.00 (-0.07)	0.00 (0.36)	-0.00 (-0.06)	0.00 (0.37)	-0.00 (-0.08)	0.00 (0.37)	-0.00 (-0.07)	0.00 (0.37)
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,168	6,168	6,168	6,168	6,168	6,168	6,168	6,168
R-squared	0.18	0.07	0.18	0.07	0.18	0.07	0.18	0.07
Coefficient Difference	0.01	0.04***	0.03	0.03*	0.01*	0.02	0.04***	0.05***

This table reports the regression of the market adjusted abnormal returns (in absolute value) and abnormal trading volume to conference calls and managers' GAI discussions, over the two-day window [0, +1], where day 0 refers to the conference call date. Panel A reports the full sample results. Panel B (C) constrains sample to firms whose retail trading is in the lowest (highest) tercile. Panel D reports the results for changes of initiative-related GAI discussions with different features. The dependent variables and key non-indicator independent variable $\Delta GAI\%$, $\Delta GAI\%-Pos-Broad$, $\Delta GAI\%-Initiative$, and $\Delta GAI\%-NonInitiative$ are standardized. Variable definitions are in Appendix D. t -statistics (in the parentheses right below the coefficient estimates) are based on standard errors clustered at the industry level. In Panel C, we also report the p -value of testing the difference of the coefficient with the corresponding coefficient in Panel B in the square parenthesis below the t -statistic. ***, **, and * indicate two-sided significance at the 1%, 5%, and 10% levels, respectively.

Table 8 Robustness Tests**Panel A: Results Based on Levels of Managers' GAI Discussions**

Dependent Variable	<i>GAI%</i>	<i>GAI%-Pos</i>	<i>GAI%-Neg</i>	<i>GAI%-Neu</i>	<i>GAI%-Initiative</i>	<i>GAI%-NonInitiative</i>
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Patents</i>	0.06** (2.28)	0.07** (2.12)	-0.01 (-0.25)	0.01 (0.77)	0.06** (2.28)	0.01 (0.45)
<i>Trade Secret</i>	0.00 (0.08)	0.00 (0.16)	-0.00 (-0.03)	-0.00 (-0.23)	-0.00 (-0.02)	0.02 (1.16)
<i>R&D</i>	0.05 (1.00)	0.05 (1.04)	-0.00 (-0.25)	-0.00 (-0.06)	0.05 (1.04)	-0.03 (-1.40)
<i>Miss R&D</i>	0.01 (0.11)	0.01 (0.30)	-0.01 (-0.67)	-0.02 (-0.63)	0.01 (0.29)	-0.06 (-1.27)
<i>CS Threat</i>	0.05** (2.26)	0.05** (2.48)	0.01 (0.93)	0.01 (0.70)	0.06** (2.51)	-0.02 (-1.05)
<i>AIIE</i>	0.07*** (2.76)	0.06*** (2.63)	0.02** (2.51)	0.02** (2.55)	0.07*** (2.66)	0.03*** (3.24)
<i>Customer</i>	0.51*** (3.89)	0.47*** (4.16)	0.12** (2.48)	0.13** (2.55)	0.50*** (3.95)	0.06 (1.64)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,192	6,192	6,192	6,192	6,192	6,192
R-squared	0.10	0.08	0.01	0.01	0.10	0.00

Panel B: Results Based on Words-Based Measures of Manager GAI Discussions

Dependent Variable	$\Delta GAI\%$	$\Delta GAI\%-Pos$	$\Delta GAI\%-Neg$	$\Delta GAI\%-Neu$	$\Delta GAI\%-Initiative$	$\Delta GAI\%-NonInitiative$
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Patents</i>	0.06** (2.33)	0.07** (2.22)	-0.01 (-0.52)	0.01 (0.74)	0.06** (2.32)	-0.00 (-0.01)
<i>Trade Secret</i>	0.00 (0.07)	0.00 (0.13)	-0.00 (-0.02)	-0.01 (-0.25)	-0.00 (-0.04)	0.02 (0.93)
<i>R&D</i>	0.05 (0.98)	0.06 (1.03)	-0.00 (-0.42)	-0.00 (-0.32)	0.06 (1.03)	-0.04 (-1.47)
<i>Miss R&D</i>	0.01 (0.13)	0.02 (0.35)	-0.02 (-0.78)	-0.02 (-0.56)	0.02 (0.33)	-0.08 (-1.47)
<i>CS Threat</i>	0.05** (2.28)	0.05** (2.50)	0.01 (0.87)	0.01 (0.57)	0.06** (2.56)	-0.04 (-1.35)
<i>AIIE</i>	0.07*** (2.87)	0.06*** (2.72)	0.02** (2.30)	0.02** (2.59)	0.07*** (2.76)	0.04*** (3.48)
<i>Customer</i>	0.51*** (3.91)	0.48*** (4.15)	0.11** (2.44)	0.13** (2.50)	0.51*** (3.96)	0.03 (1.04)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,192	6,192	6,192	6,192	6,192	6,192
R-squared	0.10	0.09	0.01	0.01	0.10	0.00

This table reports results of robustness tests. Panel A reports the regression results based on the levels of managers' GAI discussions after the launch of ChatGPT. Panel B reports the regression results using words-based measures of manager GAI discussions. The dependent variables *GAI%*, *GAI%-Pos*, *GAI%-Neg*, *GAI%-Neu*, *GAI%-Initiative*, *GAI%-NonInitiative* and key non-indicator independent variables *Patents*, *R&D*, *CS Threat*, and *AIIE* are standardized. Variable definitions are in Appendix D. *t*-statistics (in parentheses) are based on standard errors clustered at the industry level. ***, **, and * indicate two-sided significance at the 1%, 5%, and 10% levels, respectively.

Table 9 Within-Industry Analysis**Panel A: Manager Discussions**

Dependent Variable	$\Delta GAI\%$	$\Delta GAI\%-Pos$	$\Delta GAI\%-Neg$	$\Delta GAI\%-Neu$	$\Delta GAI\%-Initiative$	$\Delta GAI\%-NonInitiative$
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Patents</i>	0.05* (1.81)	0.06** (2.12)	-0.01 (-0.38)	0.02 (0.65)	0.05* (1.93)	0.01 (0.21)
<i>Trade Secret</i>	0.00 (0.20)	0.01 (0.80)	-0.01 (-0.44)	-0.01 (-0.65)	0.00 (0.06)	0.01 (0.47)
<i>R&D</i>	0.10 (1.34)	0.10 (1.31)	0.01 (1.09)	0.01* (1.97)	0.10 (1.35)	-0.02 (-0.84)
<i>Miss R&D</i>	-0.09 (-1.12)	-0.08 (-1.16)	-0.02 (-0.65)	-0.03 (-0.67)	-0.07 (-0.95)	-0.15** (-2.04)
<i>CS Threat</i>	0.03* (1.91)	0.04*** (2.93)	-0.00 (-0.29)	-0.01 (-0.66)	0.04** (2.02)	-0.03 (-1.26)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,192	6,192	6,192	6,192	6,192	6,192
R-squared	0.13	0.11	0.02	0.02	0.13	0.01

Panel B: Analyst Discussions

Dependent Variable	$\Delta GAI\%-Analyst$	$\Delta GAI\%-Pos-Analyst$	$\Delta GAI\%-Neg-Analyst$	$\Delta GAI\%-Neu-Analyst$	$\Delta GAI\%-Initiative-Analyst$	$\Delta GAI\%-NonInitiative-Analyst$
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Patents</i>	-0.01 (-0.94)	-0.01 (-0.56)	-0.01 (-0.37)	-0.00 (-0.10)	-0.02 (-1.27)	0.01 (0.58)
<i>Trade Secret</i>	-0.04 (-1.26)	-0.02 (-0.80)	-0.02 (-1.24)	-0.05 (-0.82)	-0.03 (-1.11)	-0.04 (-1.27)
<i>R&D</i>	0.09 (1.24)	0.10 (1.18)	-0.01 (-0.71)	-0.01 (-0.50)	0.09 (1.24)	0.04 (0.96)
<i>Miss R&D</i>	-0.14* (-1.68)	-0.12 (-1.28)	-0.02 (-1.52)	-0.04 (-0.83)	-0.13 (-1.50)	-0.03 (-1.36)
<i>CS Threat</i>	0.03*** (2.72)	0.04** (2.06)	-0.01 (-0.82)	-0.02 (-0.85)	0.04* (1.94)	0.00 (0.73)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,192	6,192	6,192	6,192	6,192	6,192
R-squared	0.10	0.07	0.01	0.02	0.08	0.02

This table reports the results of regressing changes in managers' discussion of GAI on our variables of interests after including industry fixed effects. Variable definitions are in Appendix D. The dependent variables $\Delta GAI\%$, $\Delta GAI\%-Pos$, $\Delta GAI\%-Neg$, $\Delta GAI\%-Neu$, $\Delta GAI\%-Initiative$, $\Delta GAI\%-NonInitiative$ and key firm-specific independent variables *Patents*, *R&D*, *CS Threat*, and *AIIE* are standardized. Panel A focuses on managers' discussions, whereas Panel B focuses on analysts' discussions. *t*-statistics (in parentheses) are based on standard errors clustered at the industry level. ***, **, and * indicate two-sided significance at the 1%, 5%, and 10% levels, respectively.

Table 10 Market Reactions to Launch of ChatGPT

Dependent Variable	<i>GPT Launch CAR</i>						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Patents</i>	0.19*** (3.15)						0.14*** (2.60)
<i>Trade Secret</i>		-0.83** (-2.25)					-0.66*** (-3.27)
<i>R&D</i>			-4.52*** (-10.09)				-3.96*** (-8.47)
<i>Miss R&D</i>			-0.11 (-0.34)				-0.04 (-0.12)
<i>CS Threat</i>				3.45** (2.25)			3.72*** (2.97)
<i>AIIE</i>					0.13 (1.04)		0.19* (1.76)
<i>Customer</i>						-0.16 (-0.28)	-0.33 (-0.77)
Observations	2,869	2,869	2,869	2,869	2,869	2,869	2,869
R-squared	0.00	0.01	0.02	0.00	0.00	0.00	0.03

This table reports the regression results of the determinants of market reactions to the launch of ChatGPT on November 30, 2022. Variable definitions are in Appendix D. *t*-statistics (in parentheses) are based on standard errors clustered at the industry level. ***, **, and * indicate two-sided significance at the 1%, 5%, and 10% levels, respectively.

Online Appendix for “Corporate Responses to Generative AI: Early Evidence from Conference Calls”

Forthcoming, Review of Accounting Studies

Figures

Figure IA-1: Changes of Analysts’ GAI Discussions across Industries

Tables

Table IA-1: Univariate Comparison of Changes in Analysts’ GAI Discussions

Table IA-2: Sentiment Analysis Based on Unnormalized Dependent Variables

Table IA-3: Sentiment Analysis Based on a Composite Measure

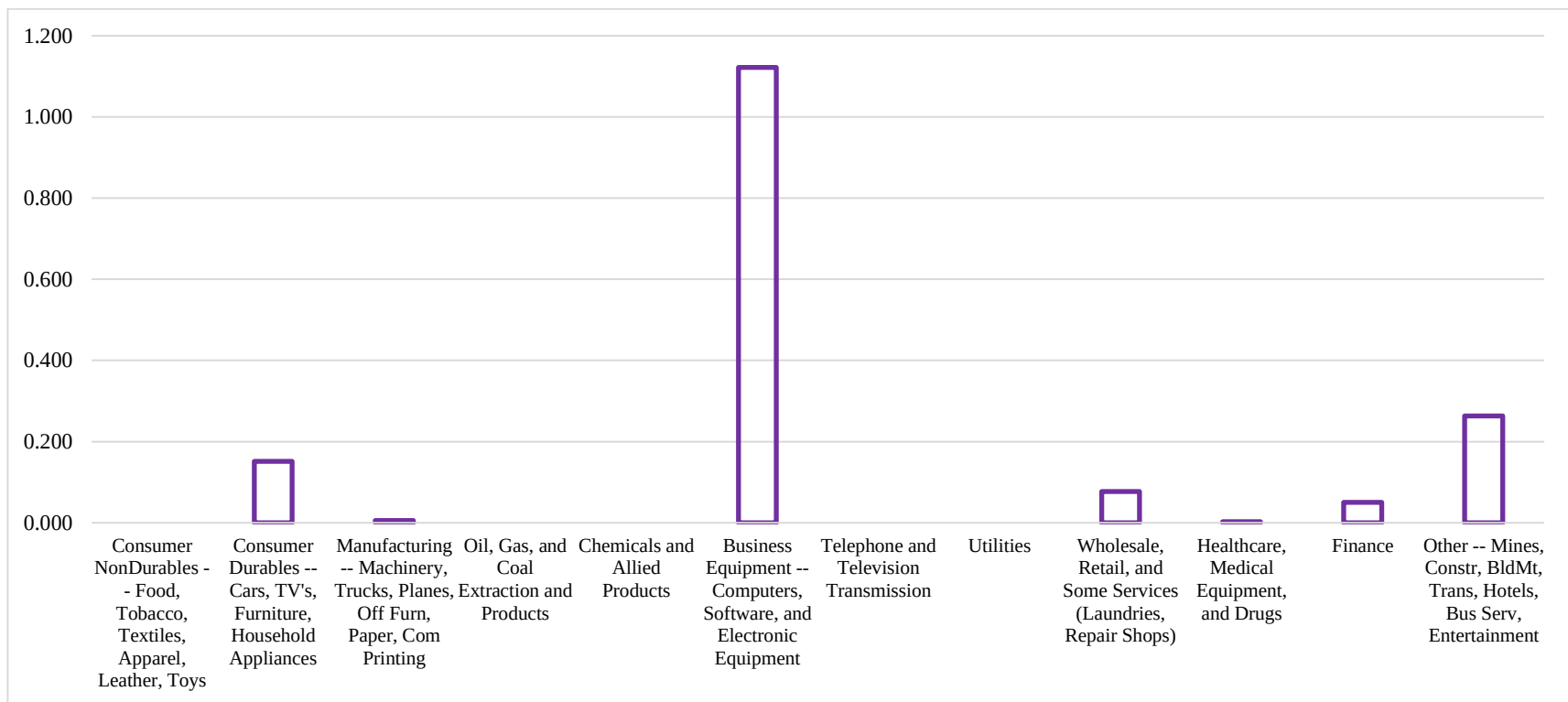


Figure IA-1A: Change of Analysts' GAI Discussion Intensity

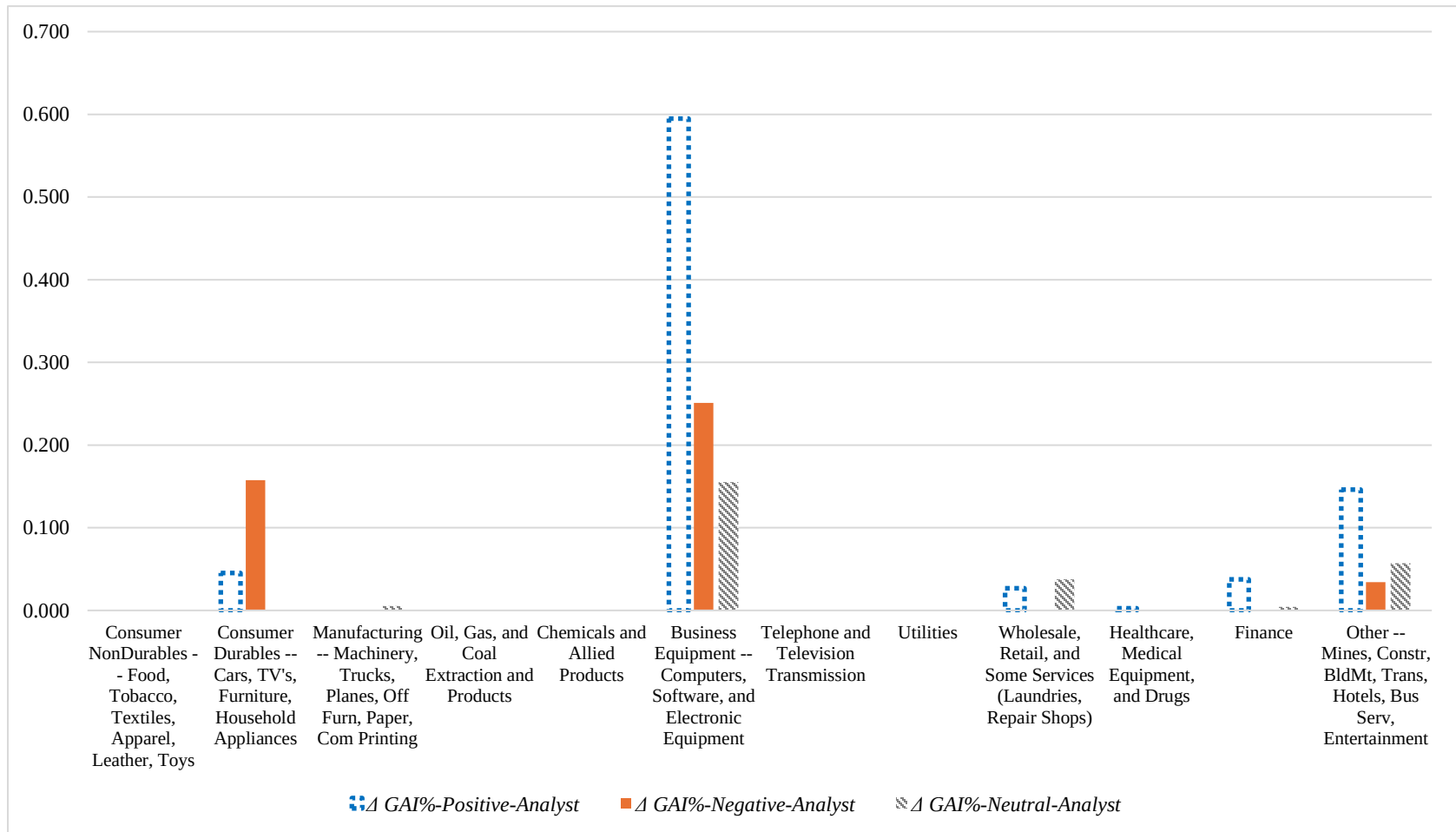


Figure IA-1B: Changes of Analysts' GAI Discussions with Positive, Negative, and Neutral Tones

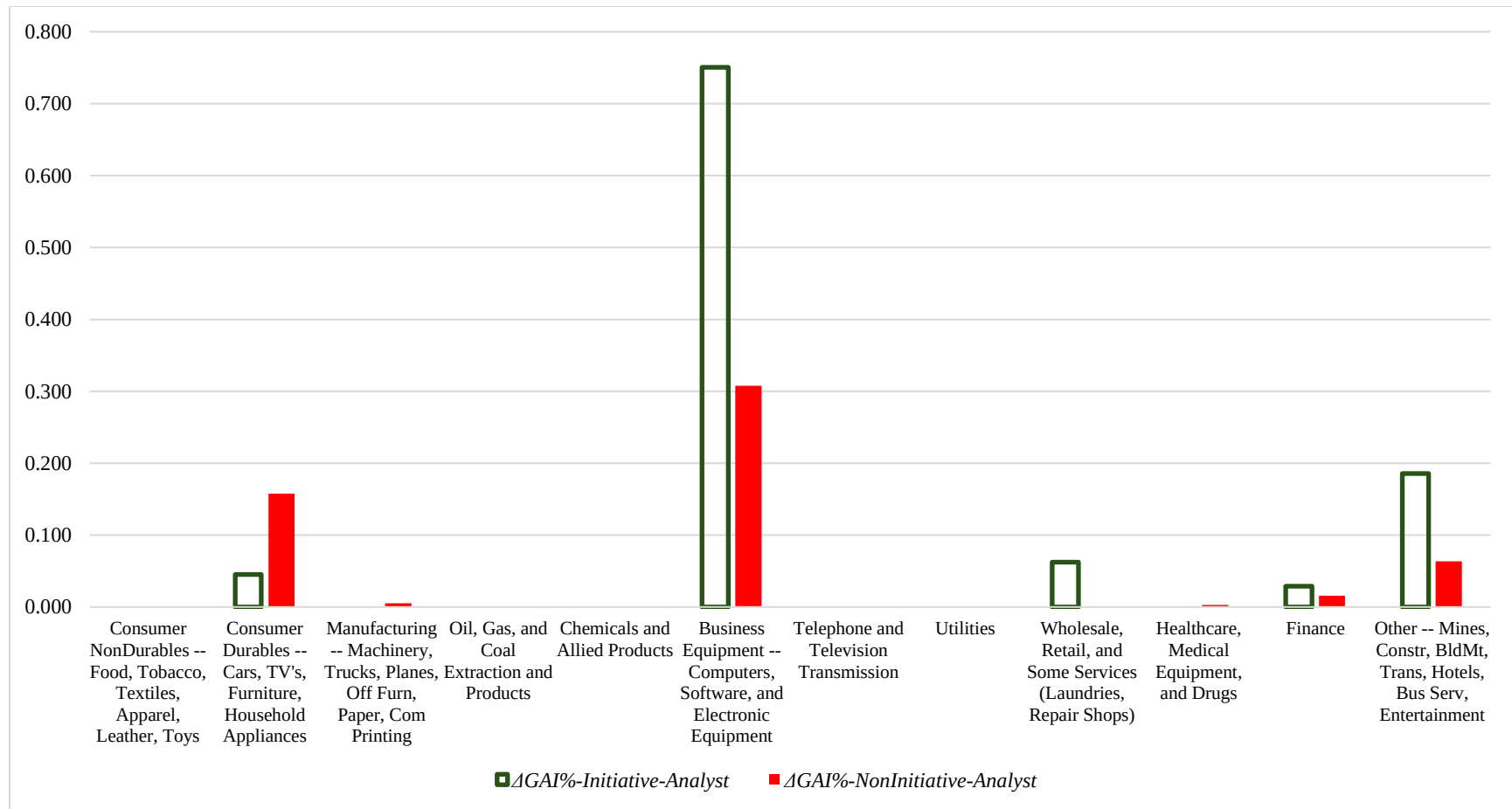


Figure IA-1C: Changes of Analysts' GAI Discussions with and without Initiatives

Figure IA-1 Changes of Analysts' GAI Discussions across Industries

Figure IA-1A presents average change of analysts' GAI discussion intensity in conference calls ($\Delta GAI\%$ -Analyst) in each of the Fama-French 12 industries. Figure IA-1B separately presents the results for positive-, negative-, and neutral-tone discussions ($\Delta GAI\%$ -Pos-Analyst, $\Delta GAI\%$ -Neg-Analyst, and $\Delta GAI\%$ -Neu-Analyst). Figure IA-1C separately presents the results for discussions with and without action initiatives ($\Delta GAI\%$ -Initiative-Analyst and $\Delta GAI\%$ -NonInitiative-Analyst). Variable definitions are in Appendix D of the manuscript.

Table IA-1 Univariate Comparison of Changes in Analysts' GAI Discussions**Panel A: Partition by High-tech Industries**

	$\Delta GAI\%-Analyst$	$\Delta GAI\%-Pos-Analyst$	$\Delta GAI\%-Neg-Analyst$	$\Delta GAI\%-Neu-Analyst$	$\Delta GAI\%-Initiative-Analyst$	$\Delta GAI\%-NonInitiative-Analyst$
High Tech Industries	0.615	0.331	0.127	0.090	0.414	0.165
Other Industries	0.037	0.014	0.014	0.008	0.020	0.016
Diff	0.578***	0.317***	0.113**	0.082***	0.394***	0.149***

Panel B: Partition by Firm Size

	$\Delta GAI\%-Analyst$	$\Delta GAI\%-Pos-Analyst$	$\Delta GAI\%-Neg-Analyst$	$\Delta GAI\%-Neu-Analyst$	$\Delta GAI\%-Initiative-Analyst$	$\Delta GAI\%-NonInitiative-Analyst$
Small Firms	0.332	0.156	0.103	0.058	0.212	0.108
Large Firms	0.303	0.180	0.034	0.038	0.211	0.069
Diff	0.029	-0.024	0.069	0.020	0.001	0.039

Panel C: Partition by Firm Age

	$\Delta GAI\%-Analyst$	$\Delta GAI\%-Pos-Analyst$	$\Delta GAI\%-Neg-Analyst$	$\Delta GAI\%-Neu-Analyst$	$\Delta GAI\%-Initiative-Analyst$	$\Delta GAI\%-NonInitiative-Analyst$
Young Firms	0.425	0.216	0.101	0.072	0.278	0.131
Mature Firms	0.197	0.114	0.033	0.020	0.137	0.041
Diff	0.228***	0.103***	0.068	0.052***	0.141***	0.090**

Panel D: Partition by Market-to-Book Ratio

	$\Delta GAI\%-Analyst$	$\Delta GAI\%-Pos-Analyst$	$\Delta GAI\%-Neg-Analyst$	$\Delta GAI\%-Neu-Analyst$	$\Delta GAI\%-Initiative-Analyst$	$\Delta GAI\%-NonInitiative-Analyst$
Growth Firms	0.520	0.281	0.112	0.082	0.342	0.137
Value Firms	0.115	0.056	0.025	0.013	0.080	0.040
Diff	0.405***	0.225***	0.087**	0.069***	0.262***	0.097***

This table reports the univariate comparison of changes of analysts' GAI discussions, overall intensity and those with different features (sentiment and initiative-relatedness) based on industry nature (high-tech industries or not) and fundamental firm characteristics (size, age, and market-to-book) partitioned based on the sample median. Variable definitions are in Appendix D of the manuscript. ***, **, and * indicate two-sided significance at the 1%, 5%, and 10% levels, respectively.

Table IA-2 Sentiment Analysis Based on Unnormalized Dependent Variables

Dependent Variable	$\Delta GAI\%-Pos$	$\Delta GAI\%-Neg$	$\Delta GAI\%-Neu$
	(1)	(2)	(3)
<i>Patents</i>	0.09** (2.14)	-0.00 (-0.24)	0.01 (0.82)
<i>Trade Secret</i>	0.01 (0.19)	-0.00 (-0.03)	-0.00 (-0.40)
<i>R&D</i>	0.08 (1.04)	-0.00 (-0.56)	-0.00 (-0.06)
<i>Miss R&D</i>	0.02 (0.29)	-0.01 (-0.68)	-0.01 (-0.74)
<i>CS Threat</i>	0.08** (2.48)	0.01 (0.87)	0.00 (0.68)
<i>AIIE</i>	0.09*** (2.71)	0.01** (2.48)	0.01** (2.43)
<i>Customer</i>	0.70*** (4.19)	0.07** (2.49)	0.04** (2.57)
Control	Yes	Yes	Yes
Observations	6,192	6,192	6,192
R-squared	0.08	0.01	0.01
Diff in Coefficients in Columns (1) and (2)			<i>p</i> -Value for Diff >0
<i>Patents</i>	0.09		0.03**
<i>Trade Secret</i>	0.01		0.41
<i>R&D</i>	0.08		0.14
<i>CS Threat</i>	0.07		<0.01***
<i>AIIE</i>	0.08		<0.01***
<i>Customer</i>	0.63		<0.01***

This table repeats the tests in Table 4, Panel A of the manuscript using unnormalized dependent variables (the dependent variables in Table 4, Panel A of the manuscript are normalized). While dependent variables are not normalized, key non-indicator independent variables, *Patents*, *R&D over Asset*, *CS Threat*, and *AIIE* are normalized. Variable definitions are in Appendix D of the manuscript. *t*-statistics (in parentheses) are based on standard errors clustered at the industry level. ***, **, and * indicate two-sided significance at the 1%, 5%, and 10% levels, respectively.

Table IA-3 Sentiment Analysis Based on a Composite Measure

Dependent Variable	$\Delta GAI\%-Pos-Broad$
	(1)
<i>Patents</i>	0.06** (2.15)
<i>Trade Secret</i>	0.00 (0.20)
<i>R&D</i>	0.05 (1.04)
<i>Miss R&D</i>	0.01 (0.29)
<i>CS Threat</i>	0.05** (2.50)
<i>AIIE</i>	0.06*** (2.74)
<i>Customer</i>	0.48*** (4.18)
Control	Yes
Observations	6,192
R-squared	0.08

This table reports a new sentiment analysis using composite measure of sentiment. $\Delta GAI\%-Pos-Broad$ is defined as $\Delta GAI\%-Pos$ minus $\Delta GAI\%-Neg$. The dependent variable and key non-indicator independent variables, *Patents*, *R&D*, *CS Threat*, and *AIIE* are normalized. Variable definitions are in Appendix D of the manuscript. *t*-statistics (in parentheses) are based on standard errors clustered at the industry level. ***, **, and * indicate two-sided significance at the 1%, 5%, and 10% levels, respectively.